

BFS Traversal

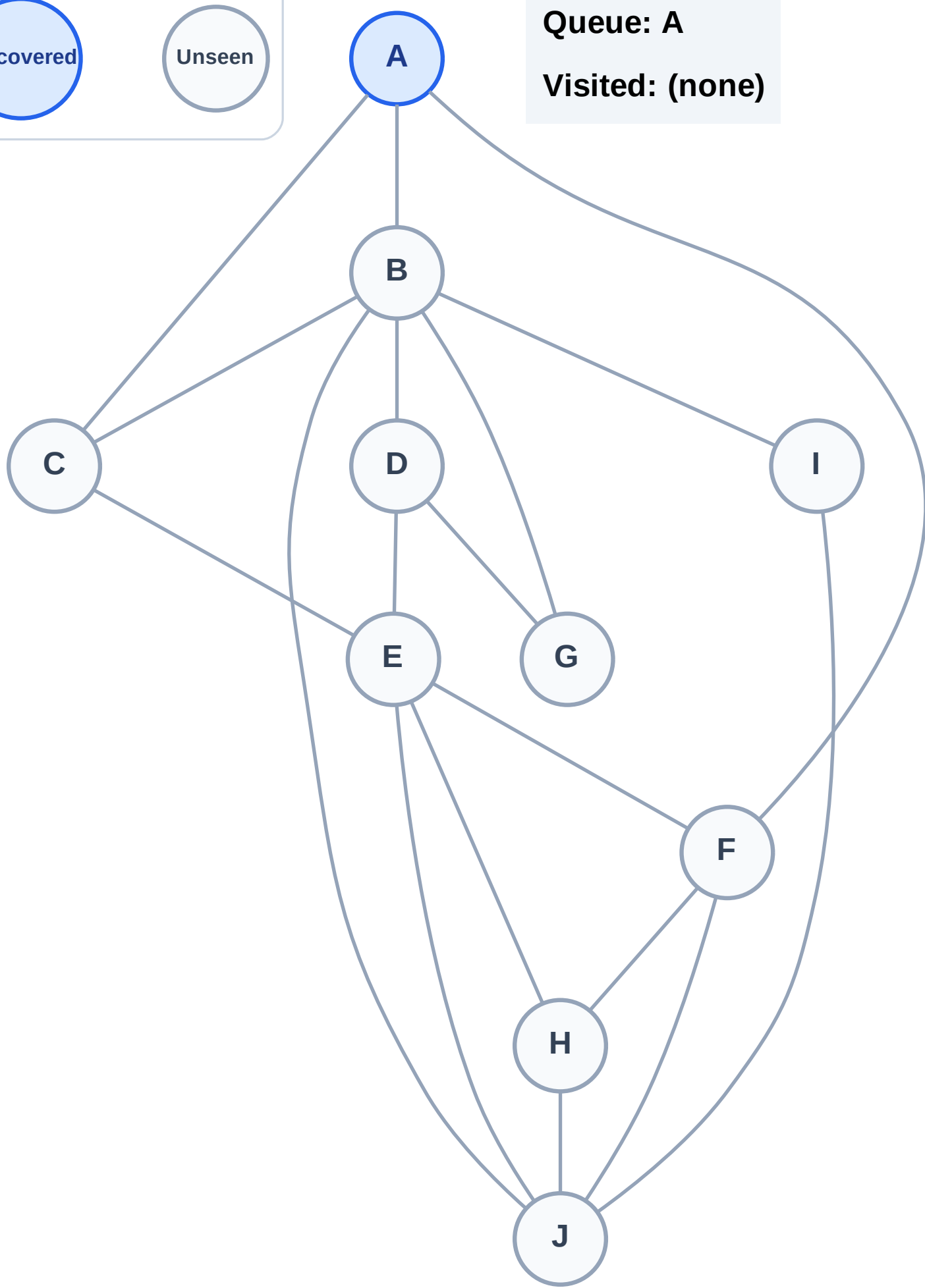
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

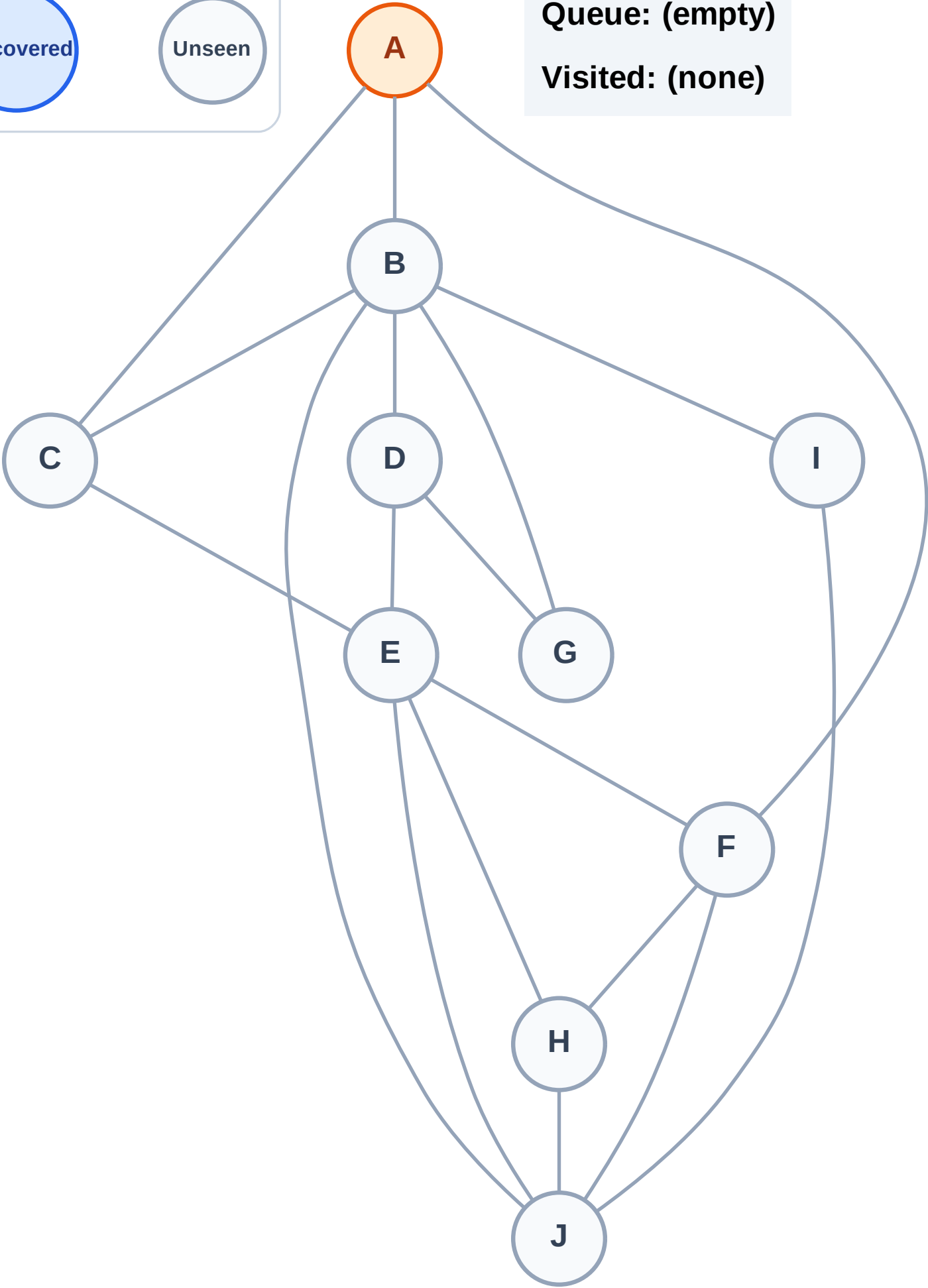
Complete

Processing

Discovered

Unseen

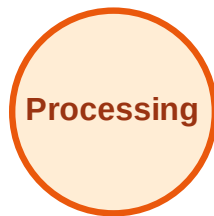
Queue: (empty)
Visited: (none)



BFS Traversal

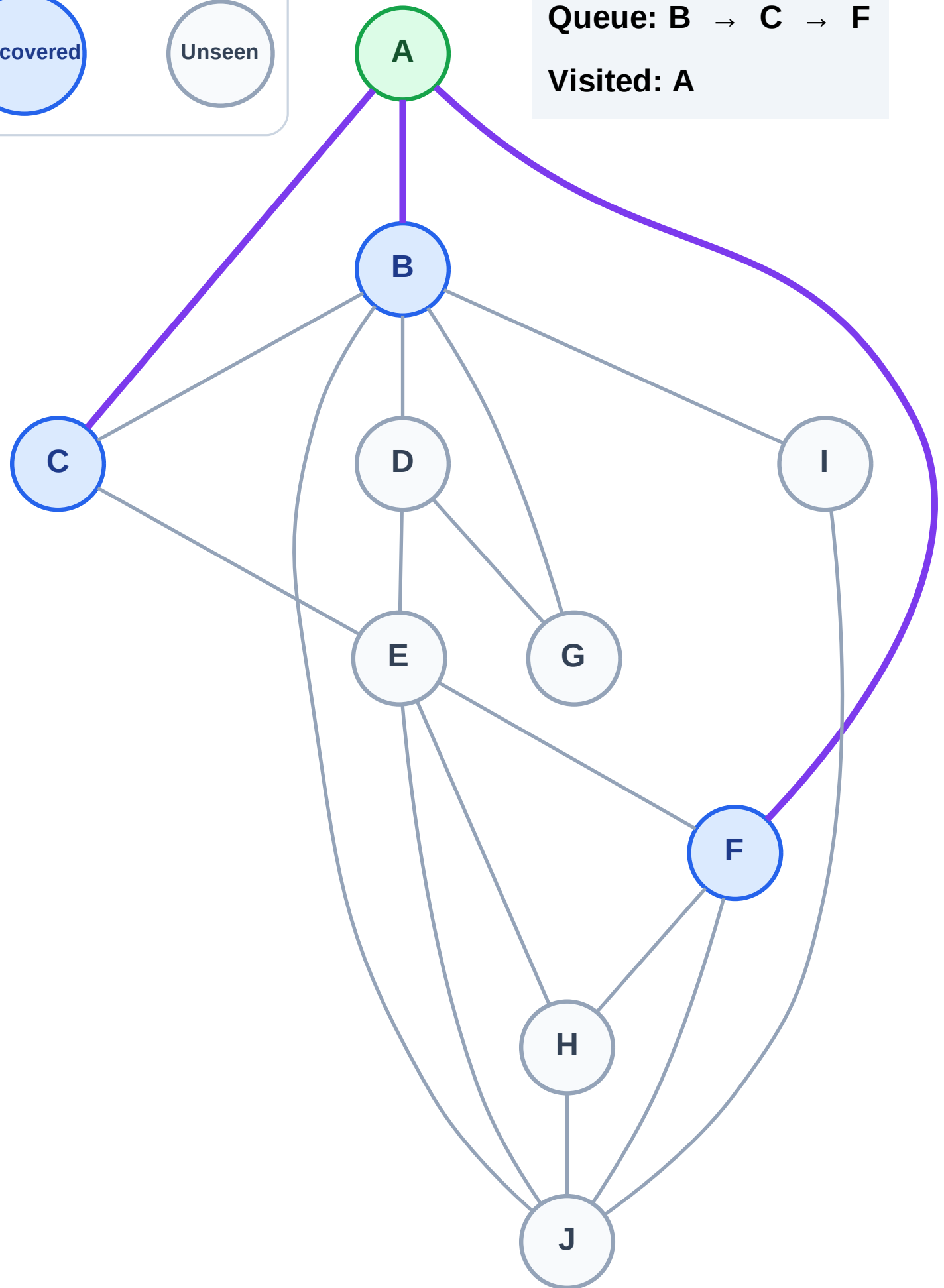
Step 3: Discover B, C, F from A

Legend



Queue: B → C → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

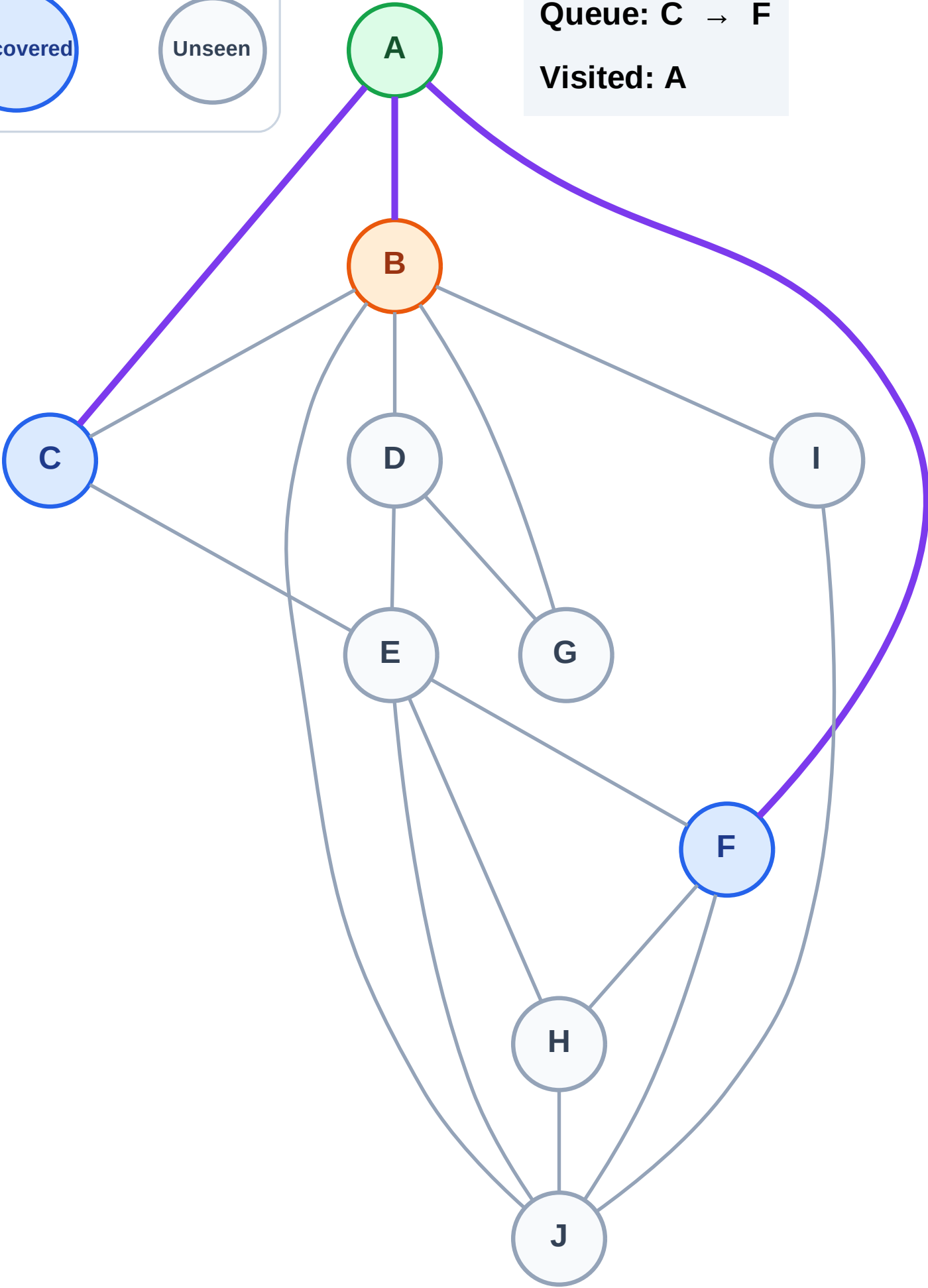
Processing

Discovered

Unseen

Queue: C → F

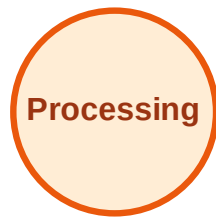
Visited: A



BFS Traversal

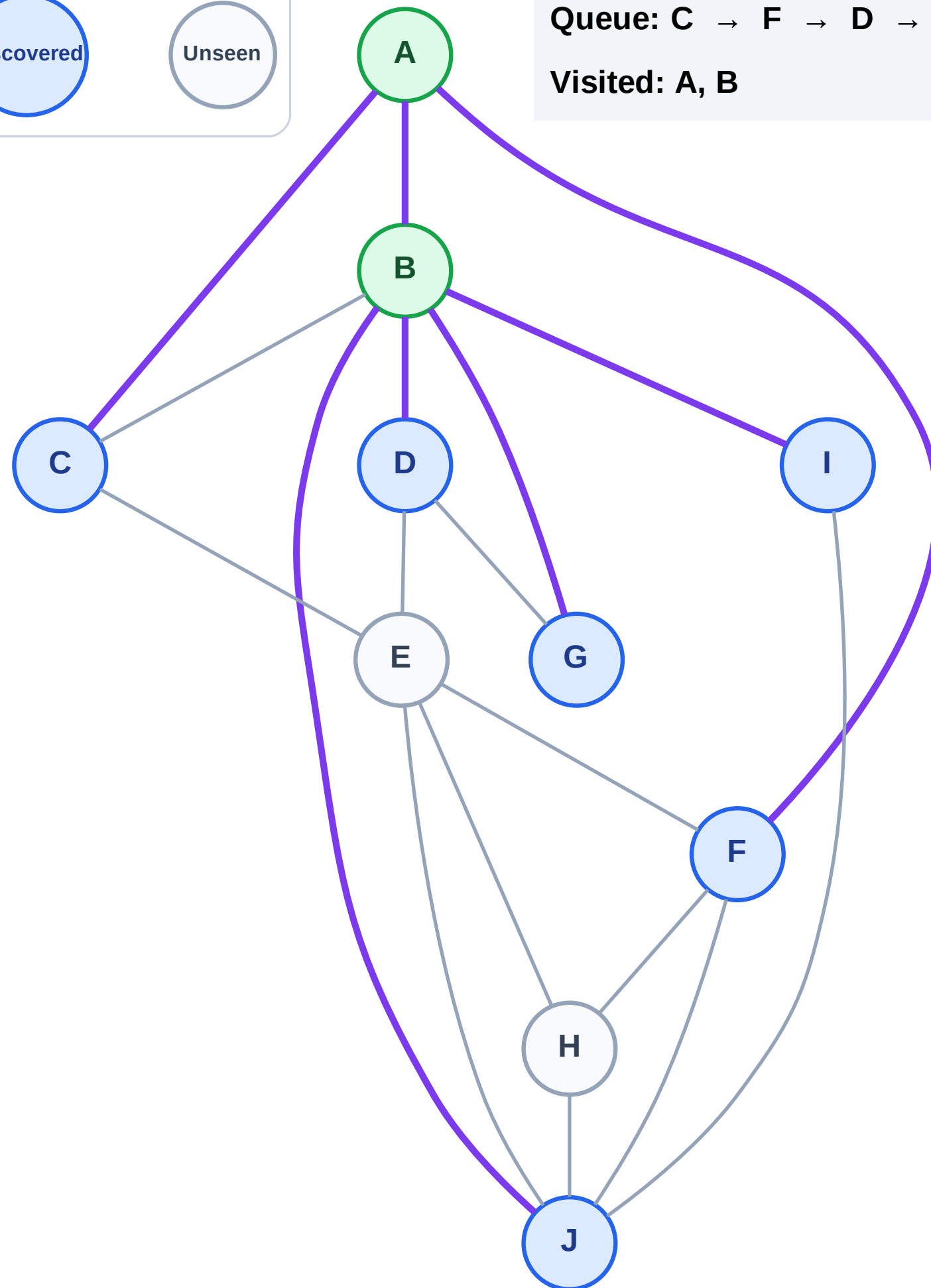
Step 5: Discover D, G, I, J from B

Legend



Queue: C → F → D → G → I → J

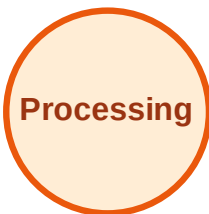
Visited: A, B



BFS Traversal

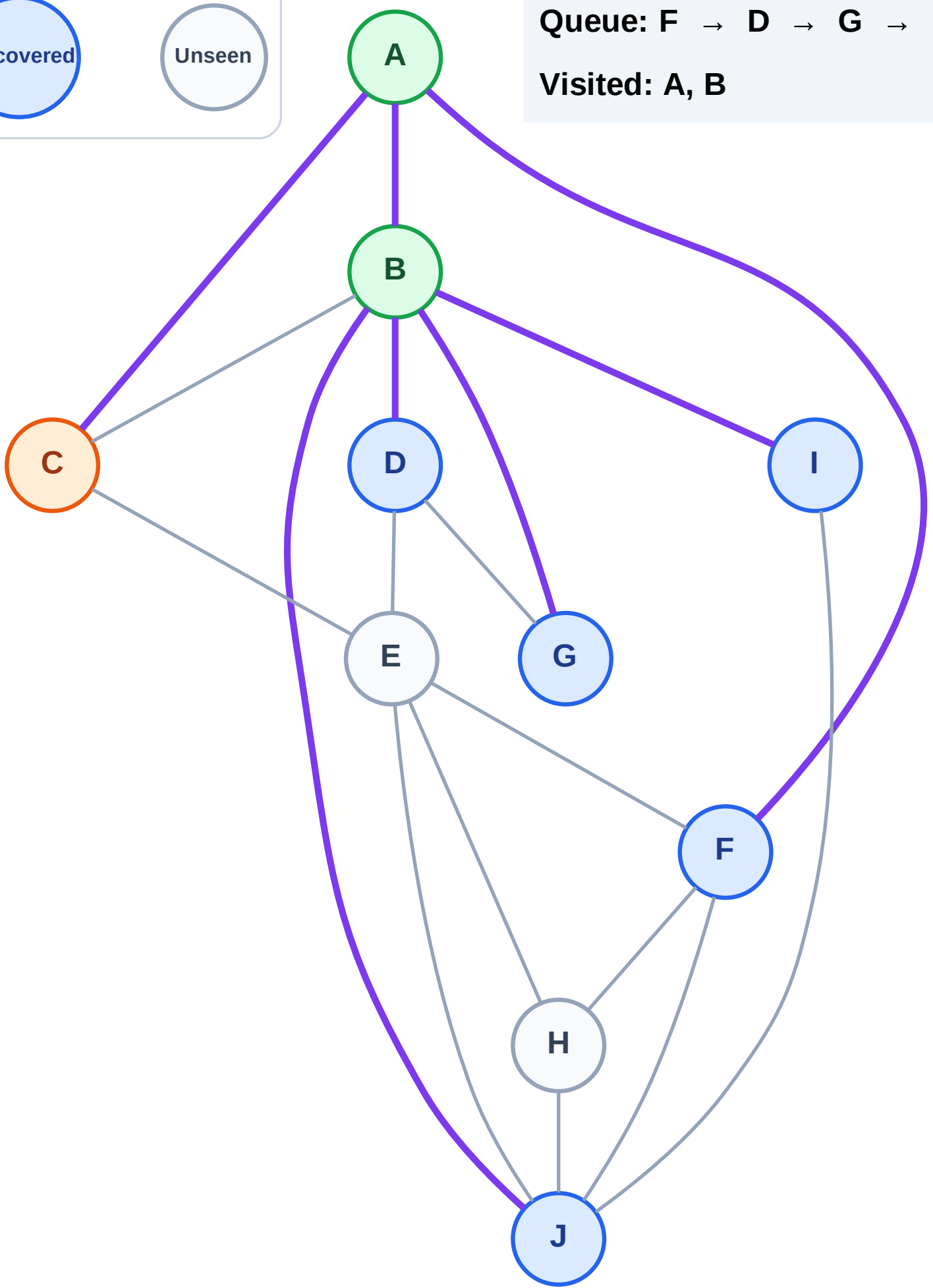
Step 6: Process node C

Legend



Queue: F → D → G → I → J

Visited: A, B



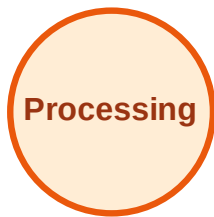
BFS Traversal

Step 7: Discover E from C

Legend



Complete



Processing



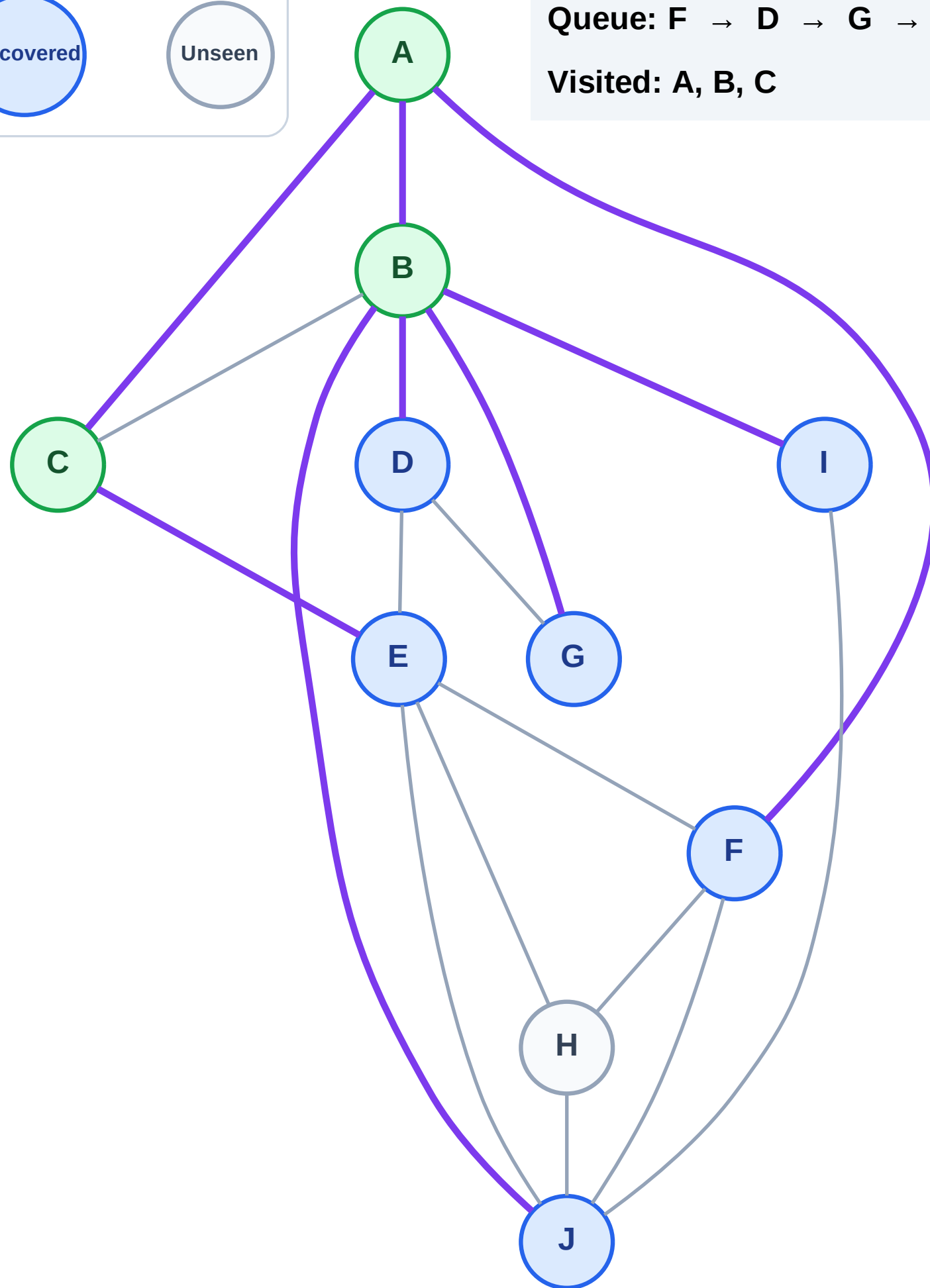
Discovered



Unseen

Queue: F → D → G → I → J → E

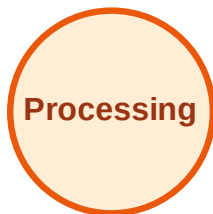
Visited: A, B, C



BFS Traversal

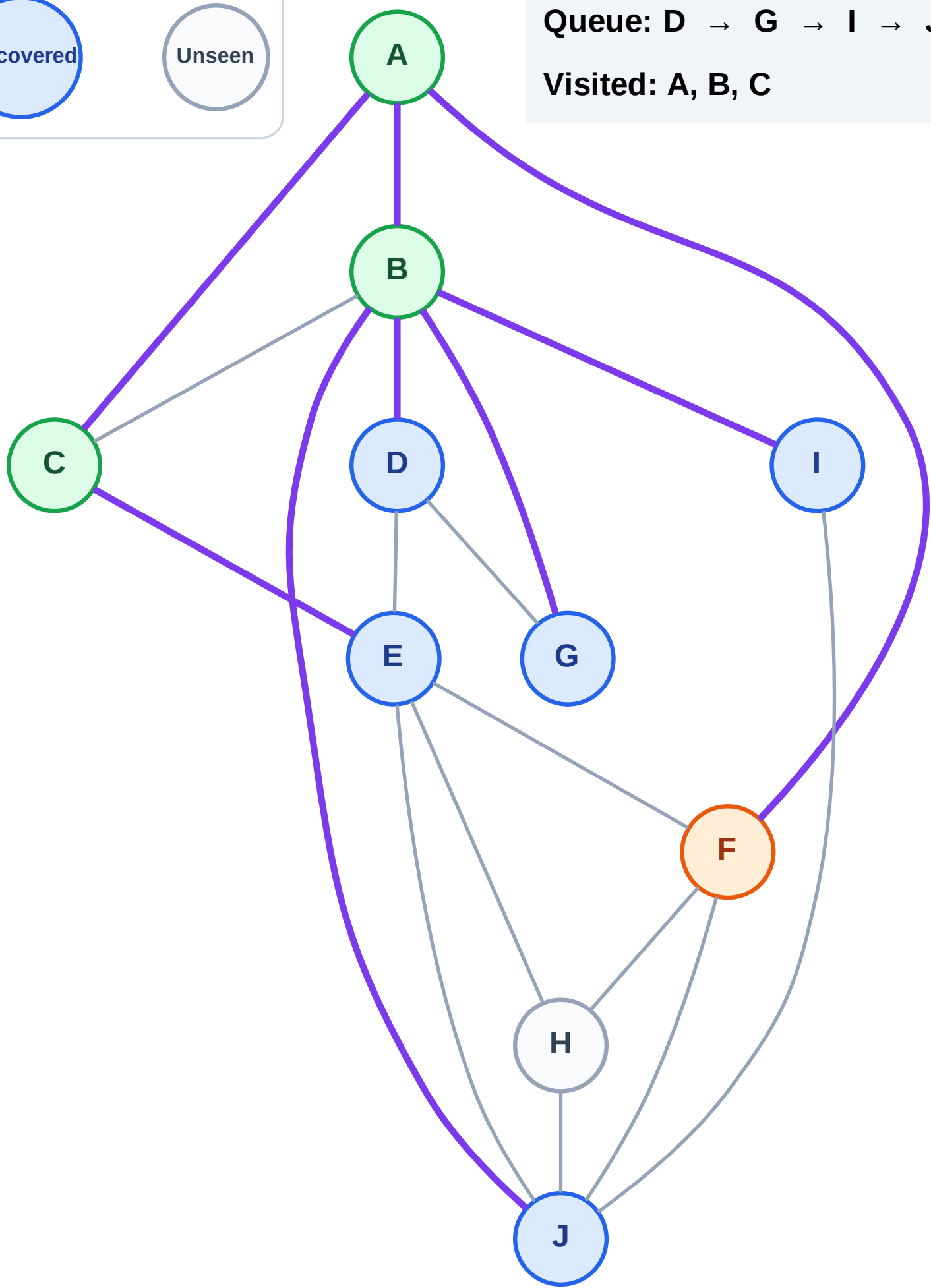
Step 8: Process node F

Legend



Queue: D → G → I → J → E

Visited: A, B, C



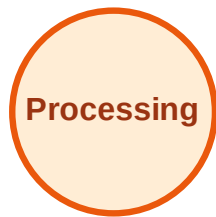
BFS Traversal

Step 9: Discover H from F

Legend



Complete



Processing



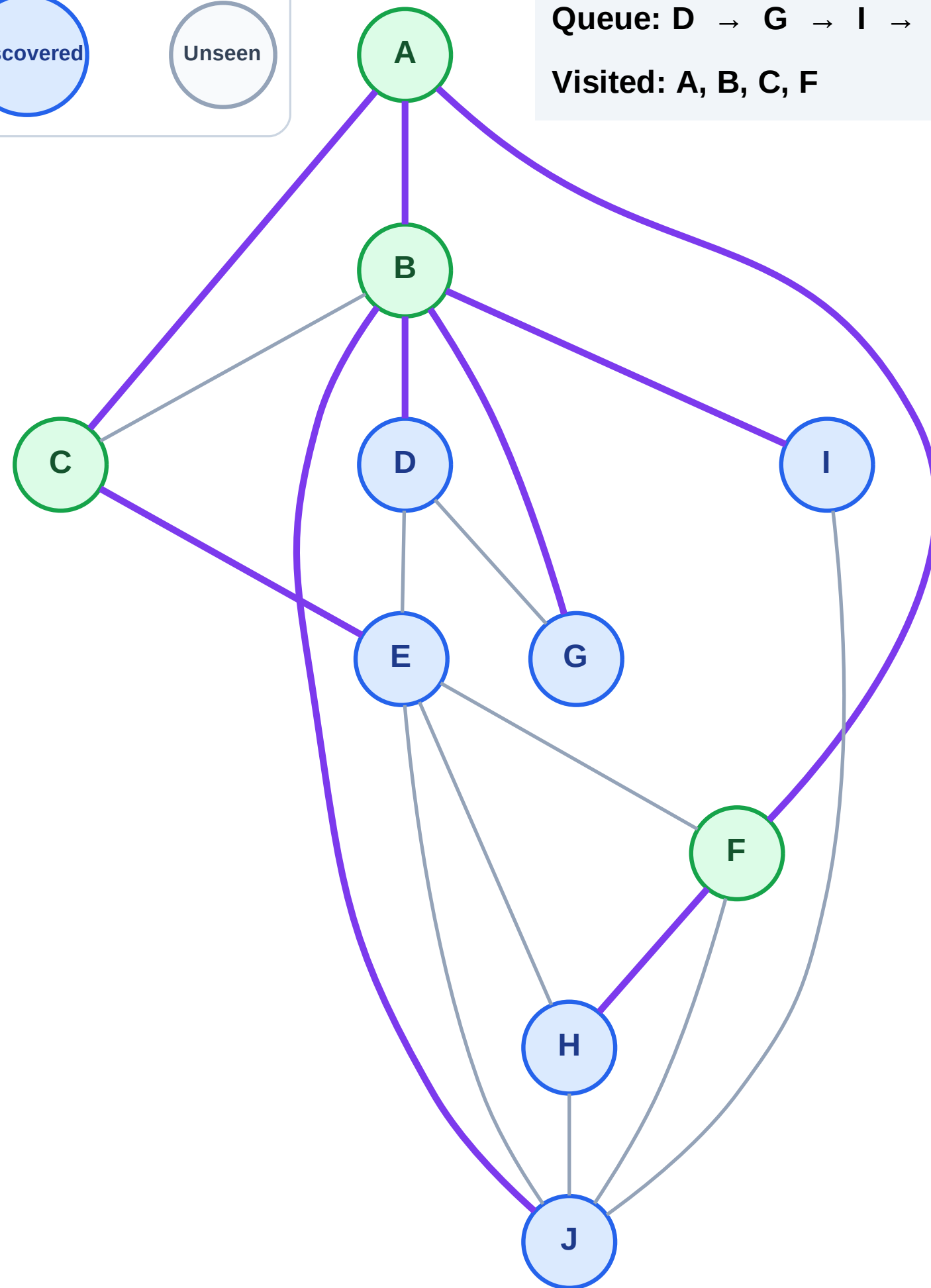
Discovered



Unseen

Queue: D → G → I → J → E → H

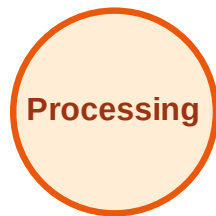
Visited: A, B, C, F



BFS Traversal

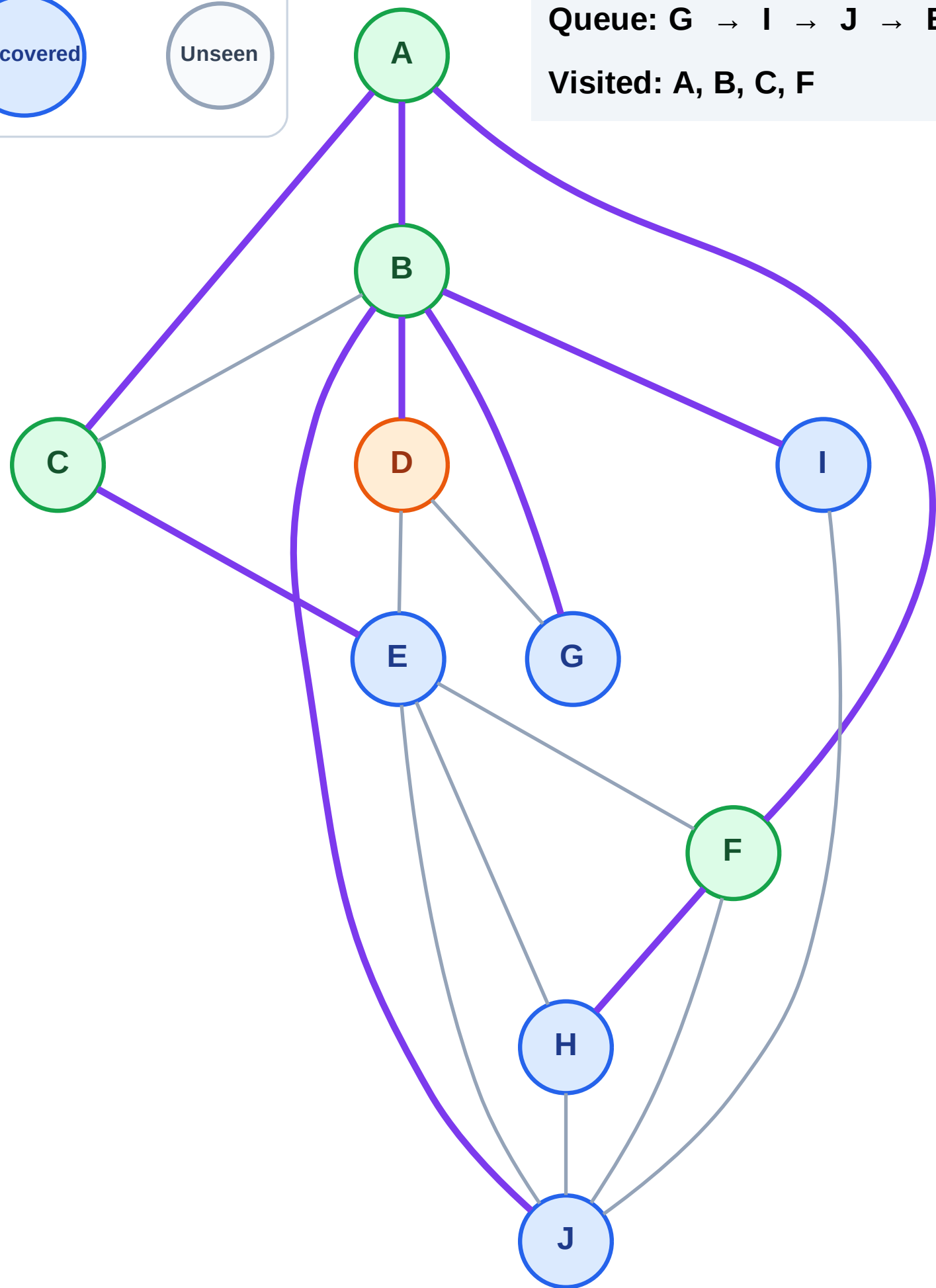
Step 10: Process node D

Legend



Queue: G → I → J → E → H

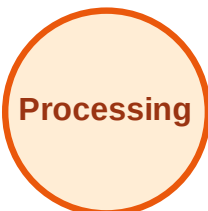
Visited: A, B, C, F



BFS Traversal

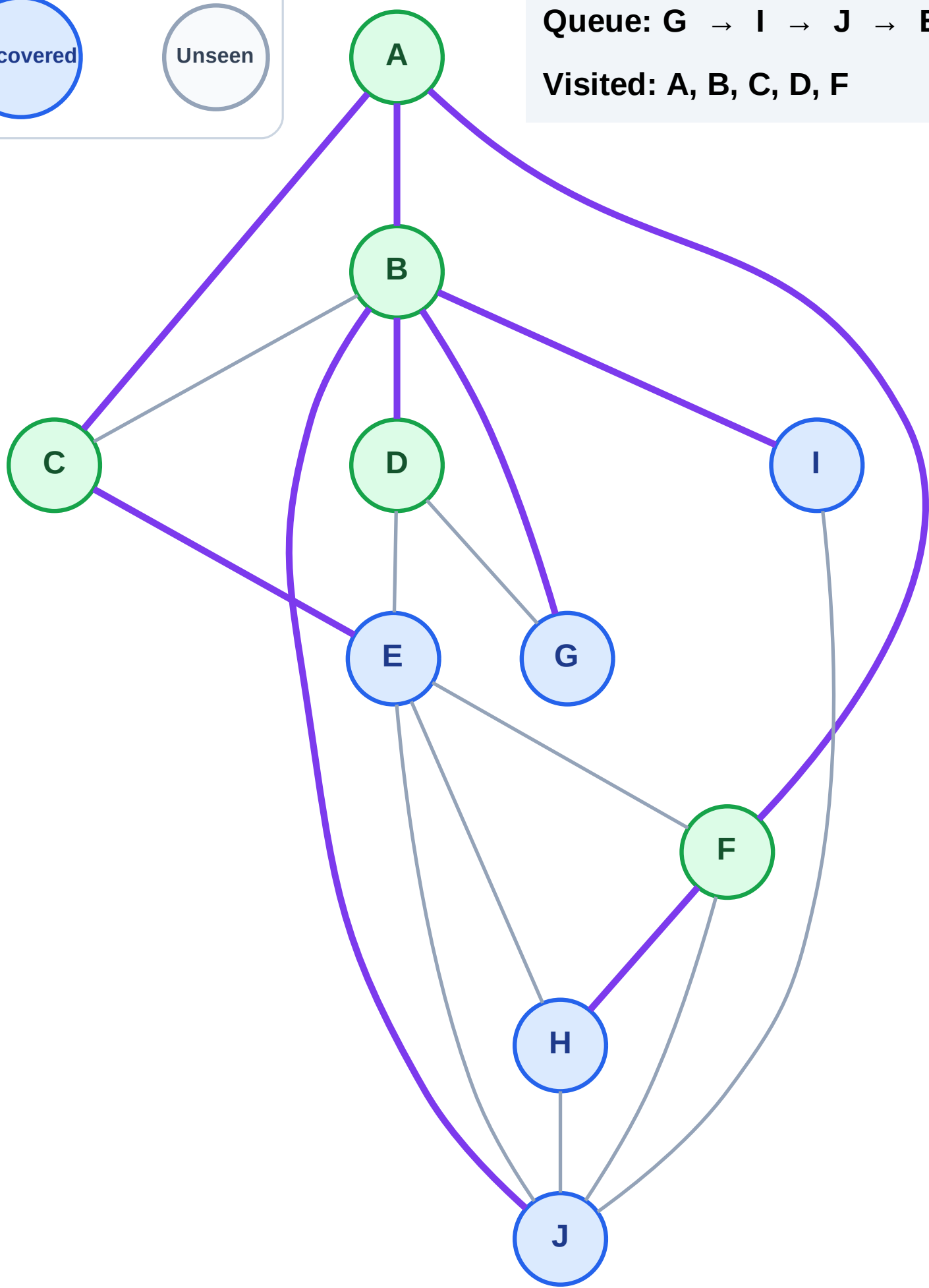
Step 11: No new neighbors from D

Legend



Queue: G → I → J → E → H

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node G

Legend

Complete

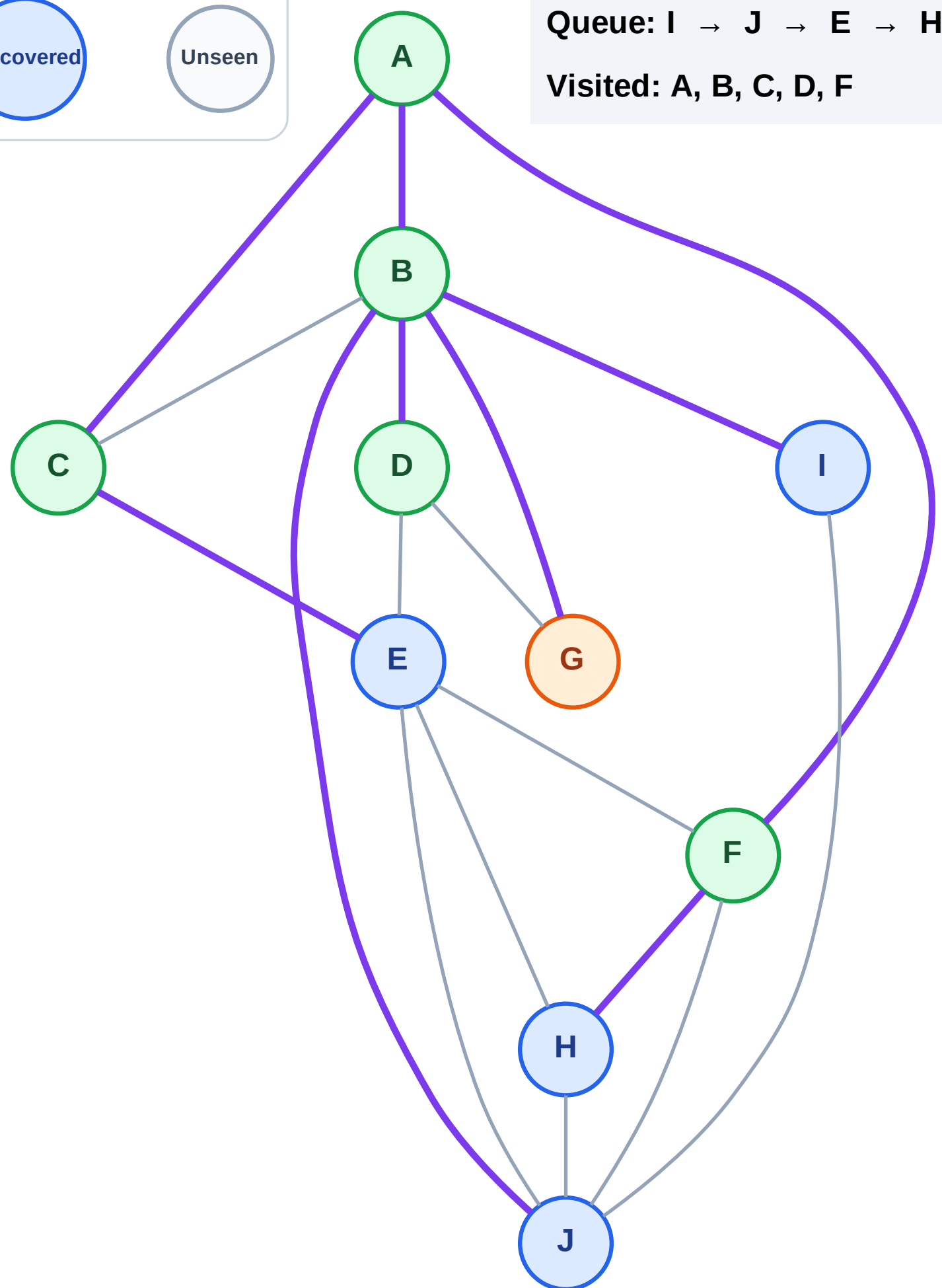
Processing

Discovered

Unseen

Queue: I → J → E → H

Visited: A, B, C, D, F



BFS Traversal

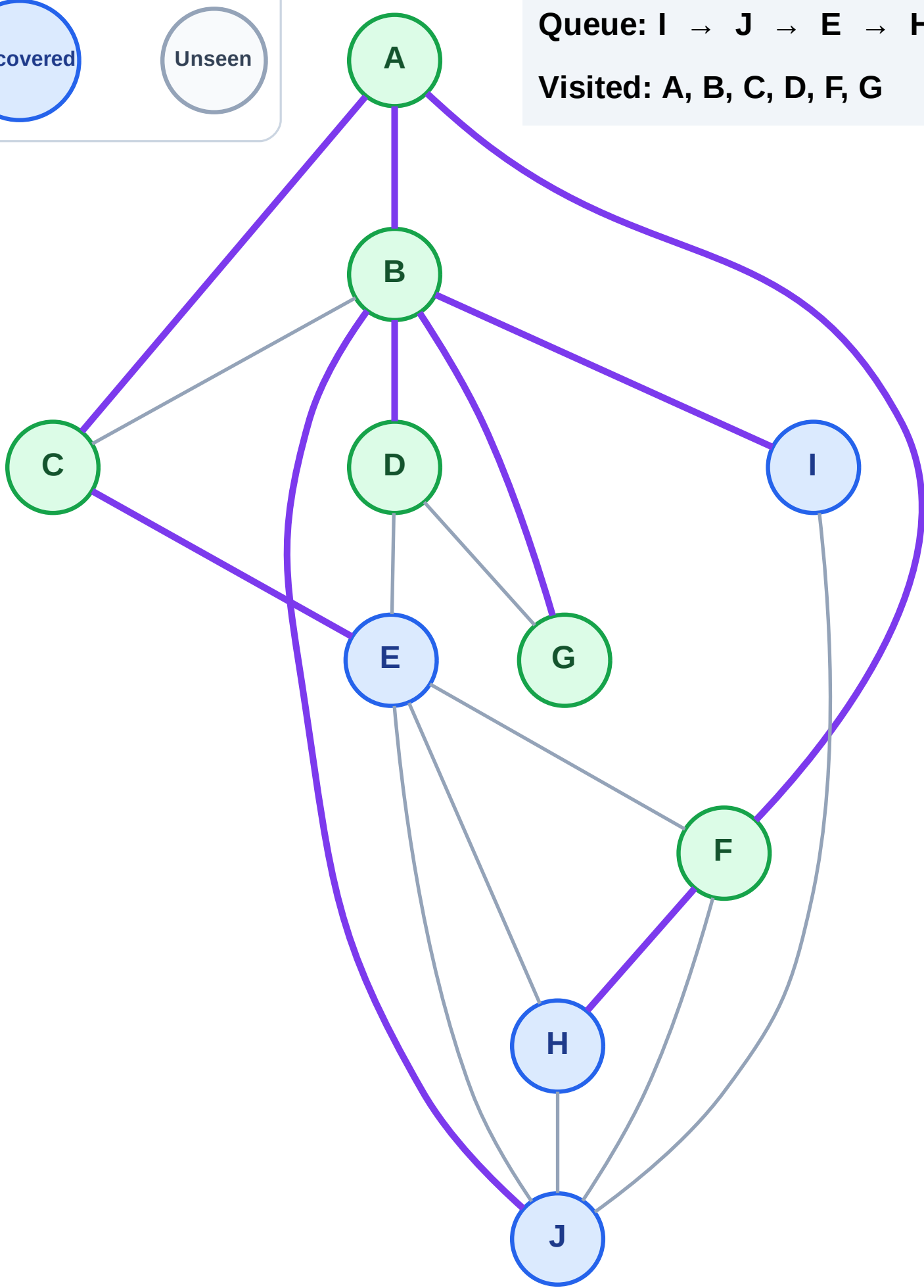
Step 13: No new neighbors from G

Legend



Queue: I → J → E → H

Visited: A, B, C, D, F, G



BFS Traversal

Step 14: Process node I

Legend

Complete

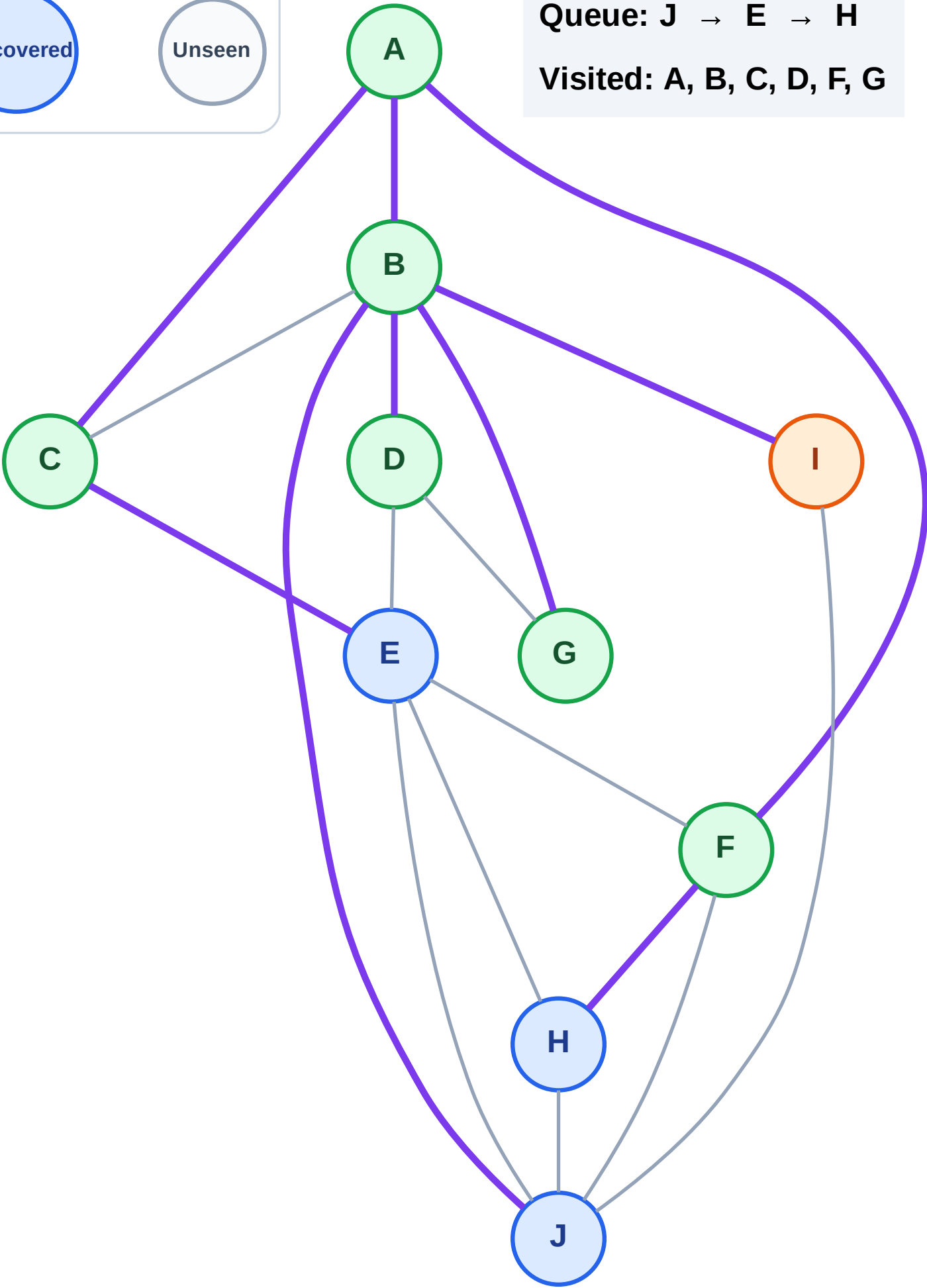
Processing

Discovered

Unseen

Queue: J → E → H

Visited: A, B, C, D, F, G



BFS Traversal

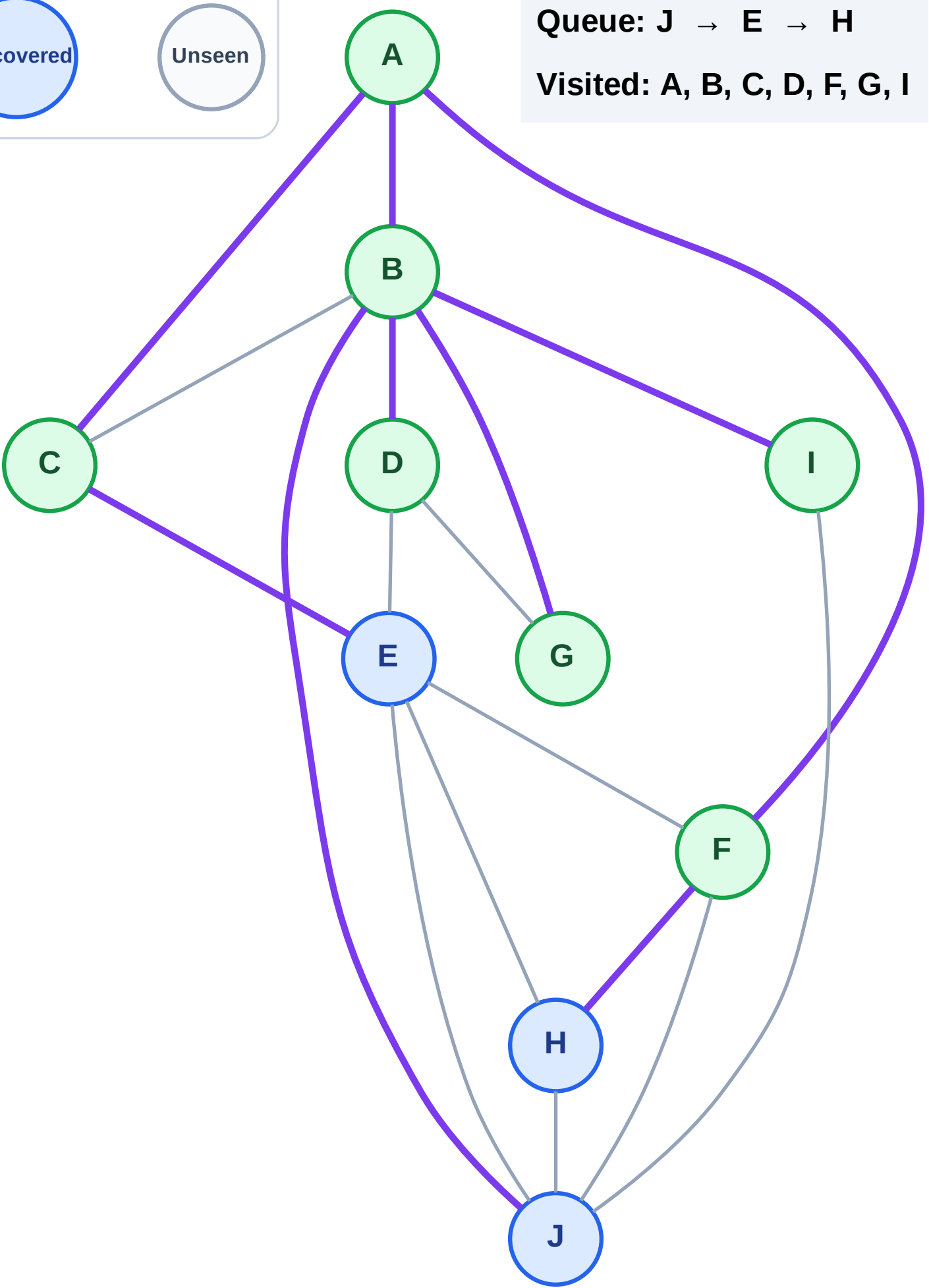
Step 15: No new neighbors from I

Legend



Queue: J → E → H

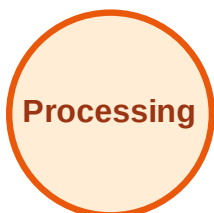
Visited: A, B, C, D, F, G, I



BFS Traversal

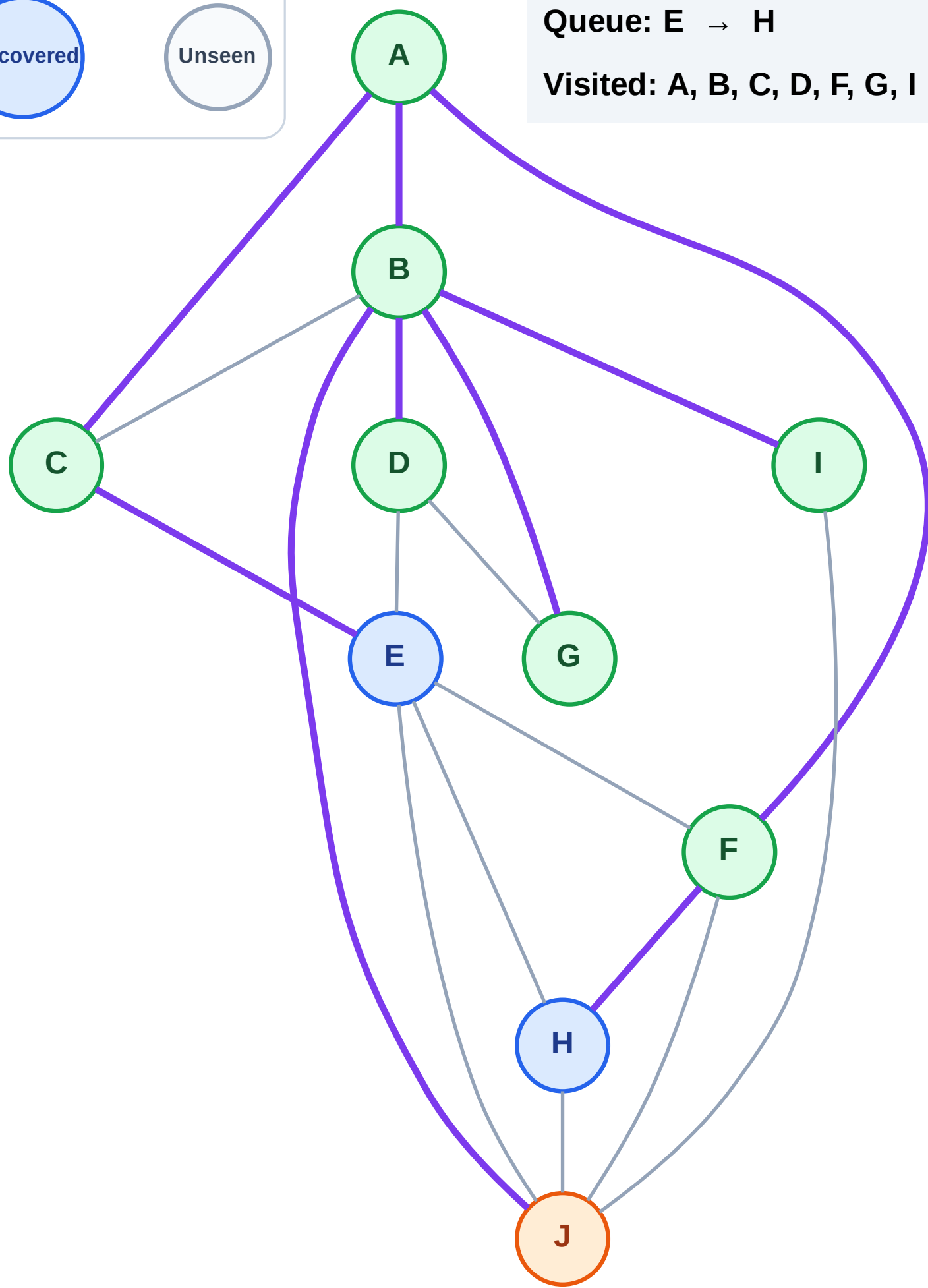
Step 16: Process node J

Legend



Queue: E → H

Visited: A, B, C, D, F, G, I



BFS Traversal

Step 17: No new neighbors from J

Legend

Complete

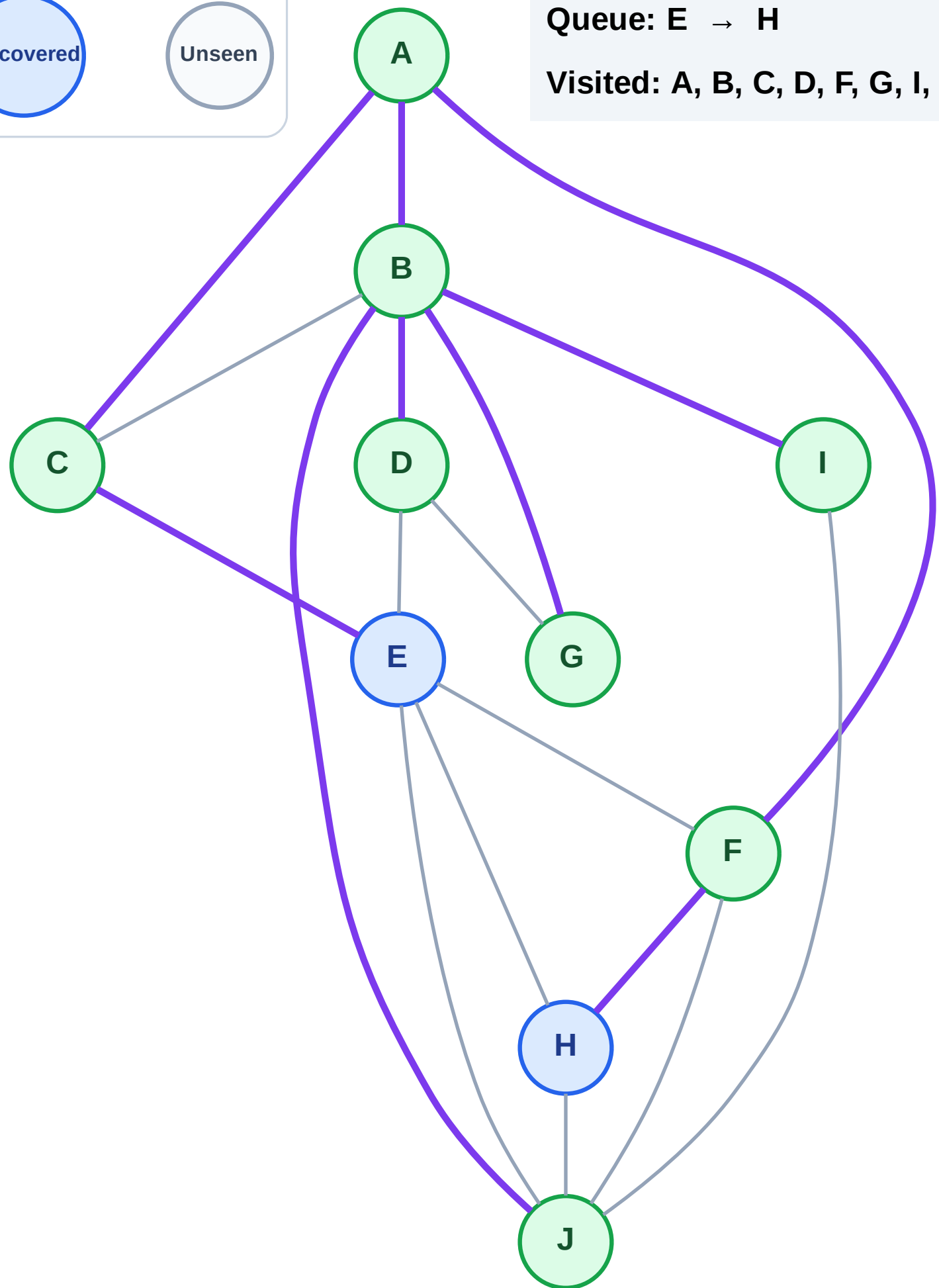
Processing

Discovered

Unseen

Queue: E → H

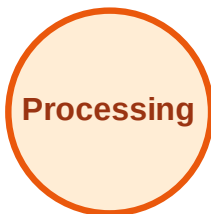
Visited: A, B, C, D, F, G, I, J



BFS Traversal

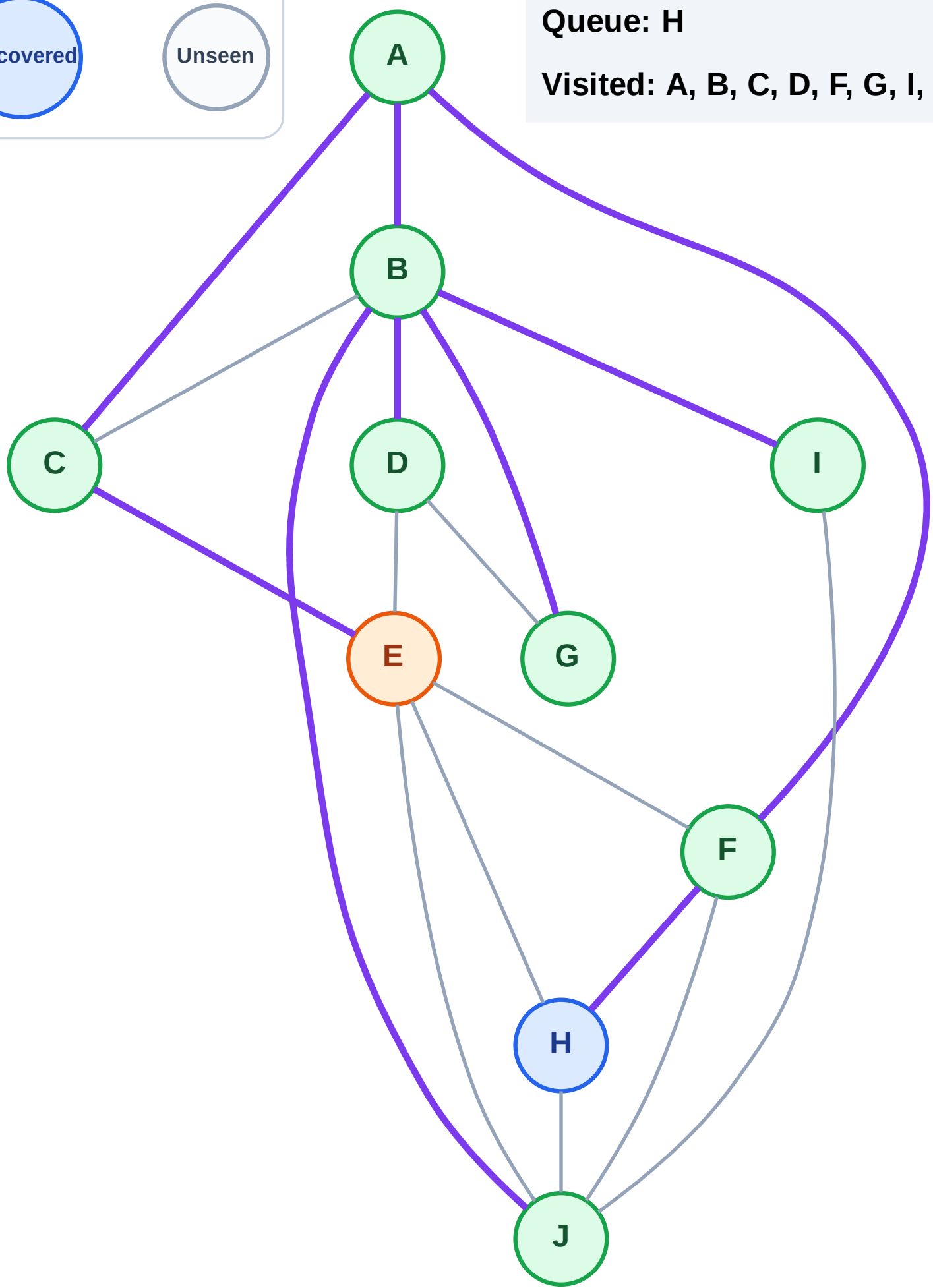
Step 18: Process node E

Legend



Queue: H

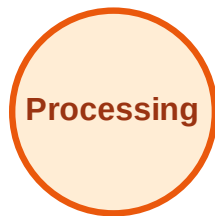
Visited: A, B, C, D, F, G, I, J



BFS Traversal

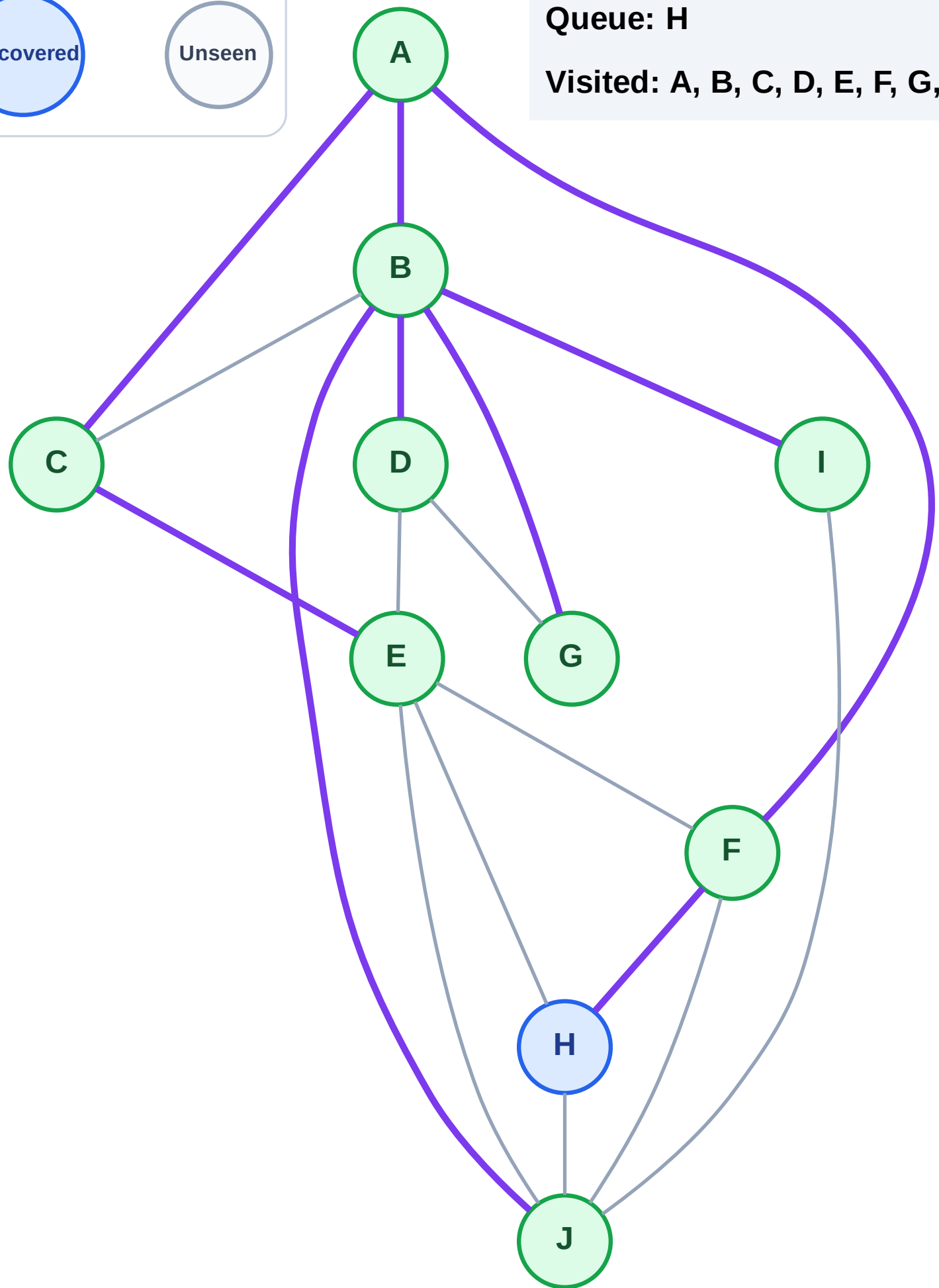
Step 19: No new neighbors from E

Legend

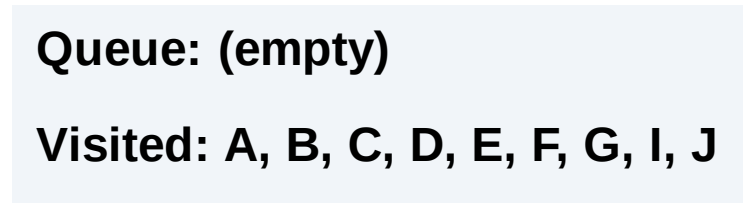


Queue: H

Visited: A, B, C, D, E, F, G, I, J

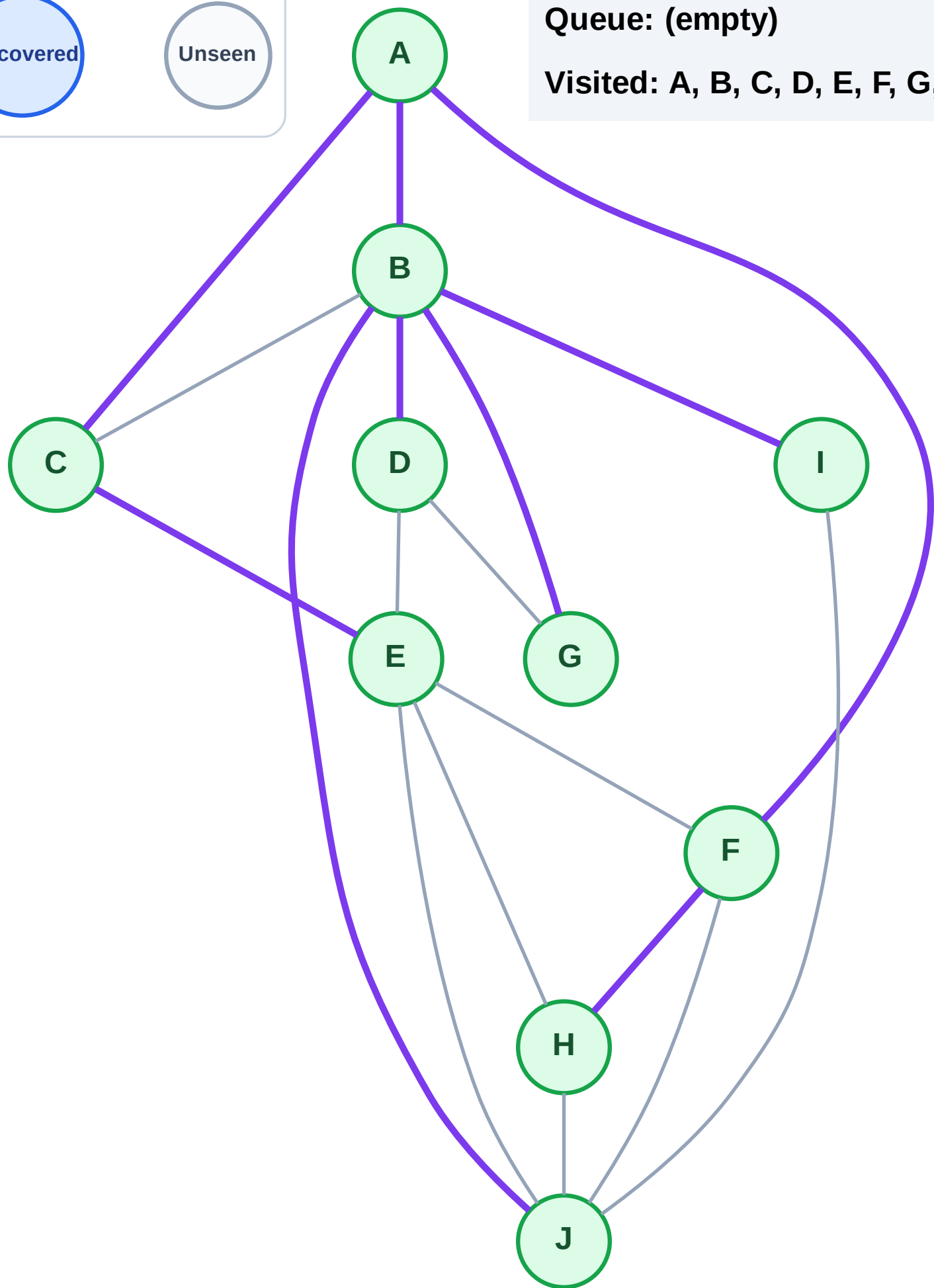
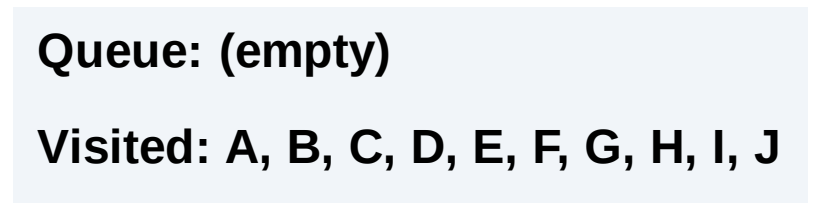


Step 20: Process node H



Step 21: No new neighbors from H

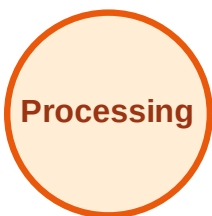
Step 21: No new neighbors from H



DFS Traversal

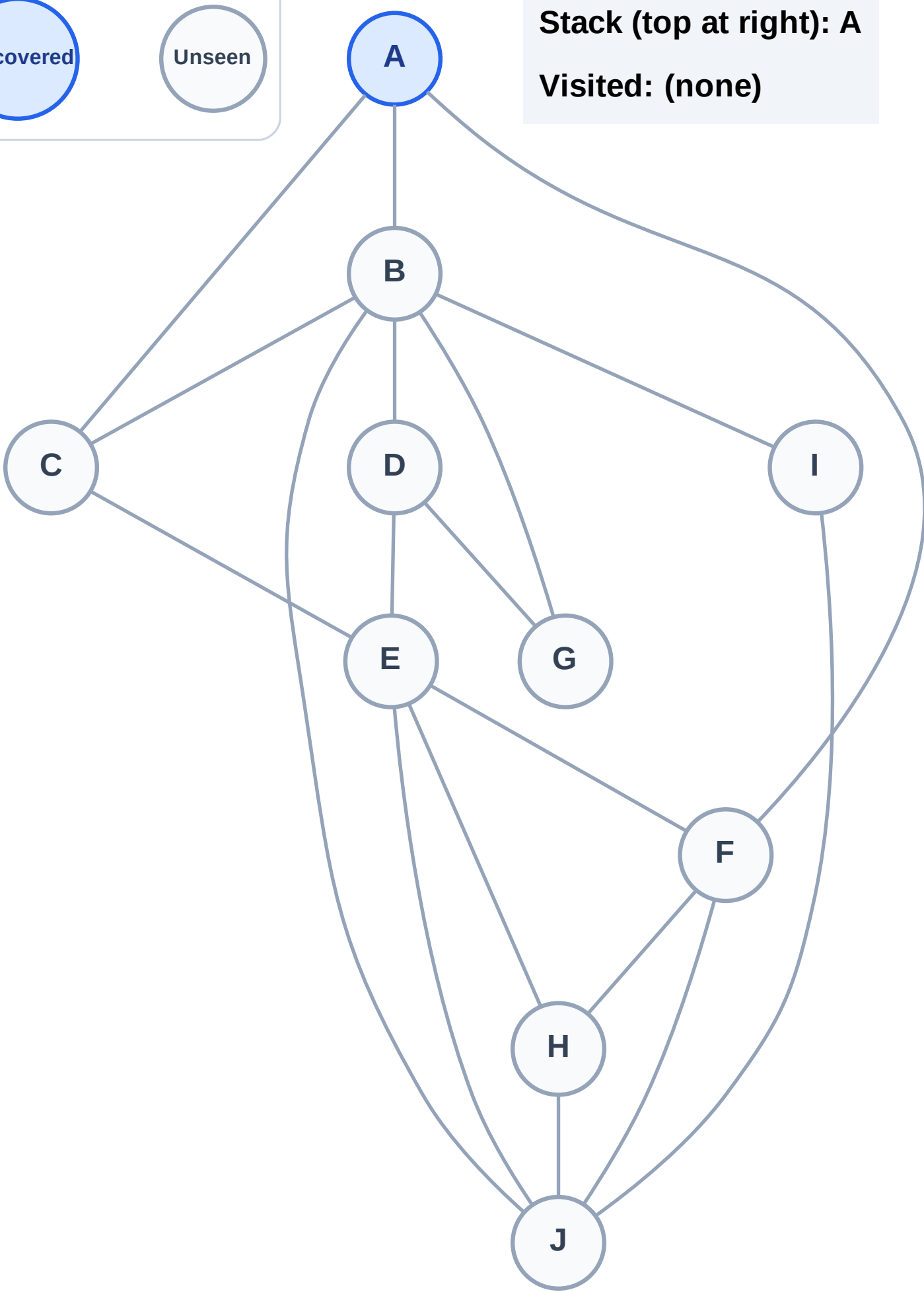
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

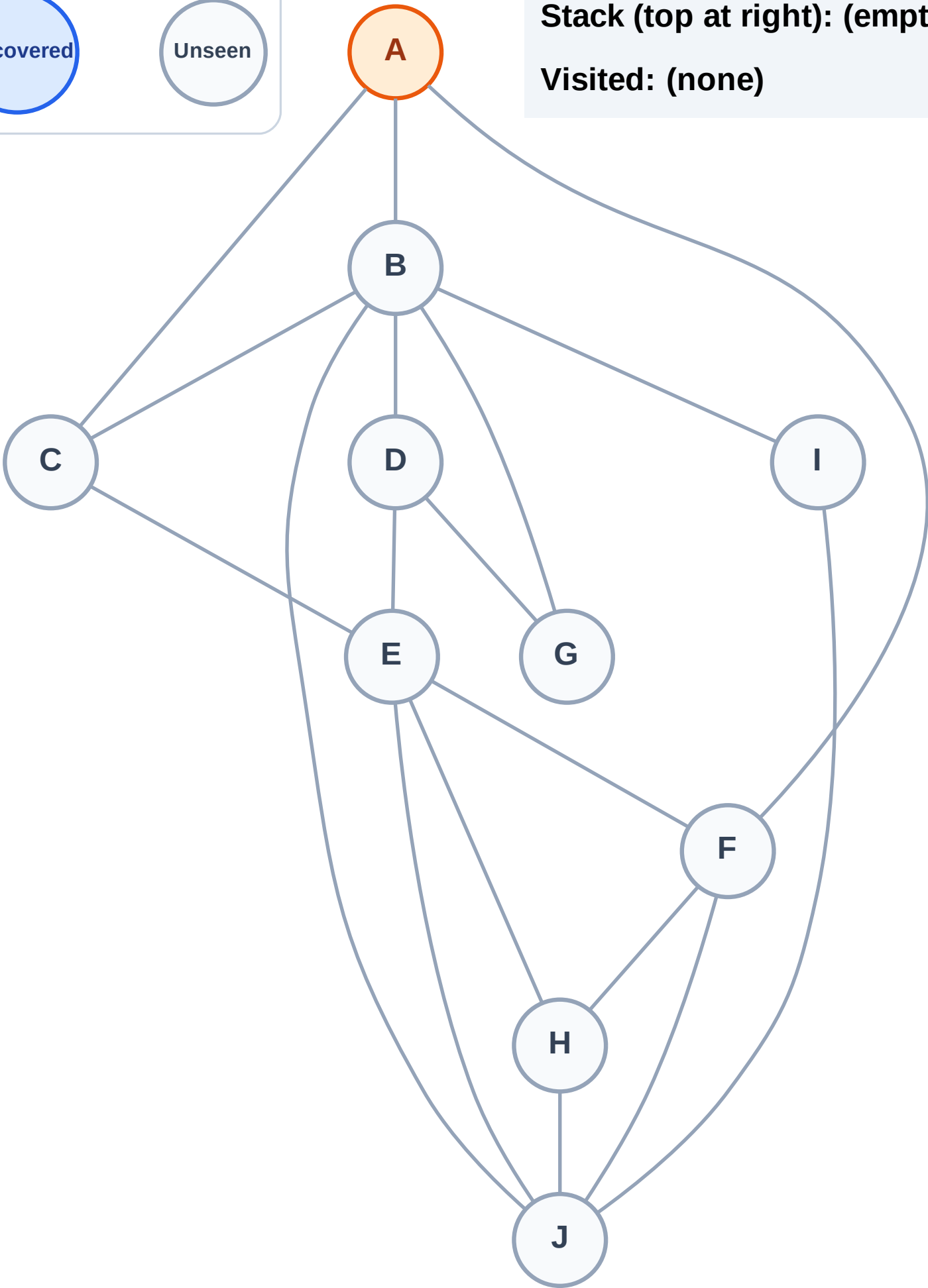
Complete

Processing

Discovered

Unseen

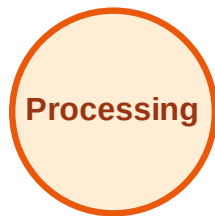
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

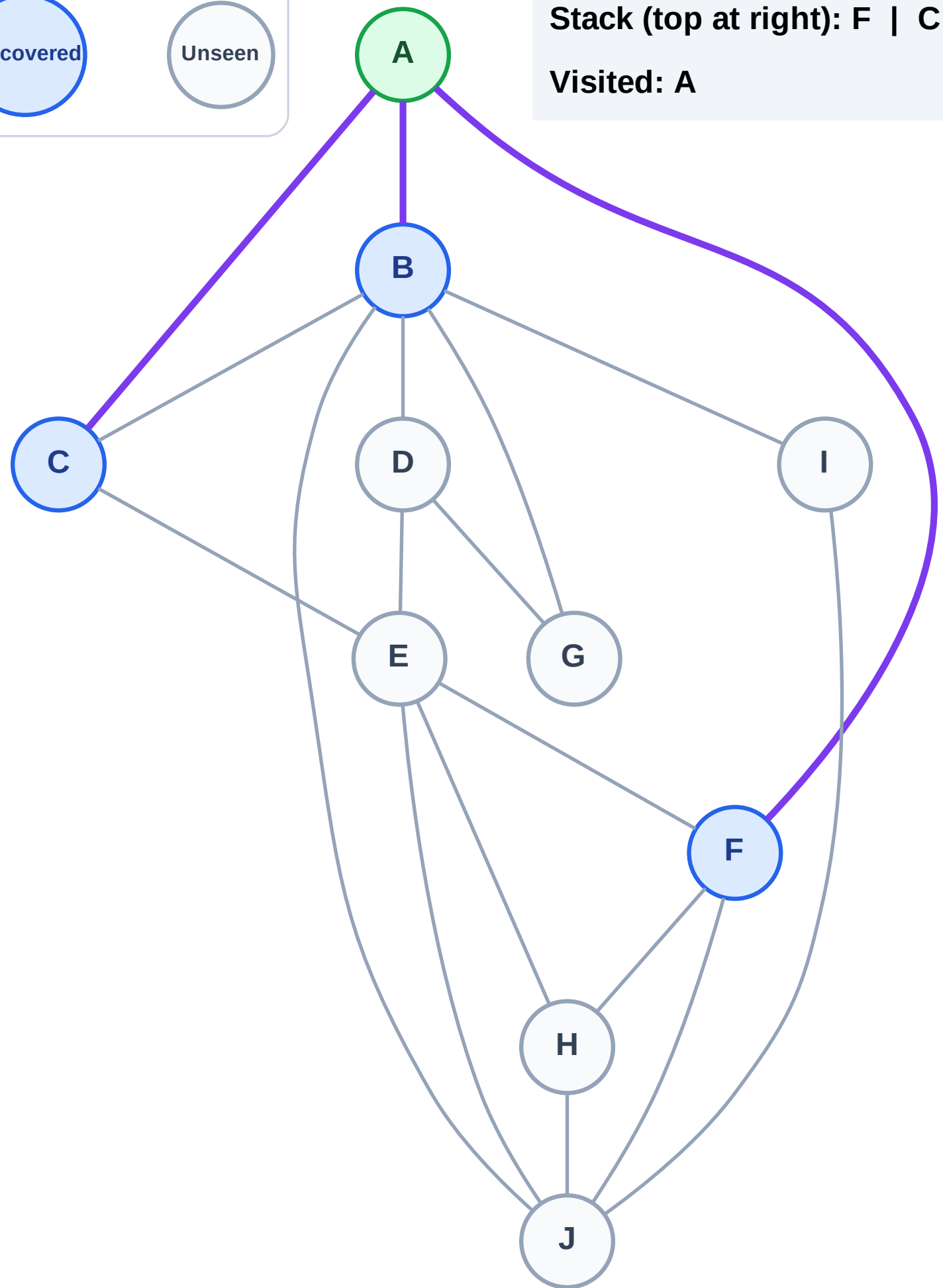
Step 3: Discover F, C, B from A

Legend



Stack (top at right): F | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

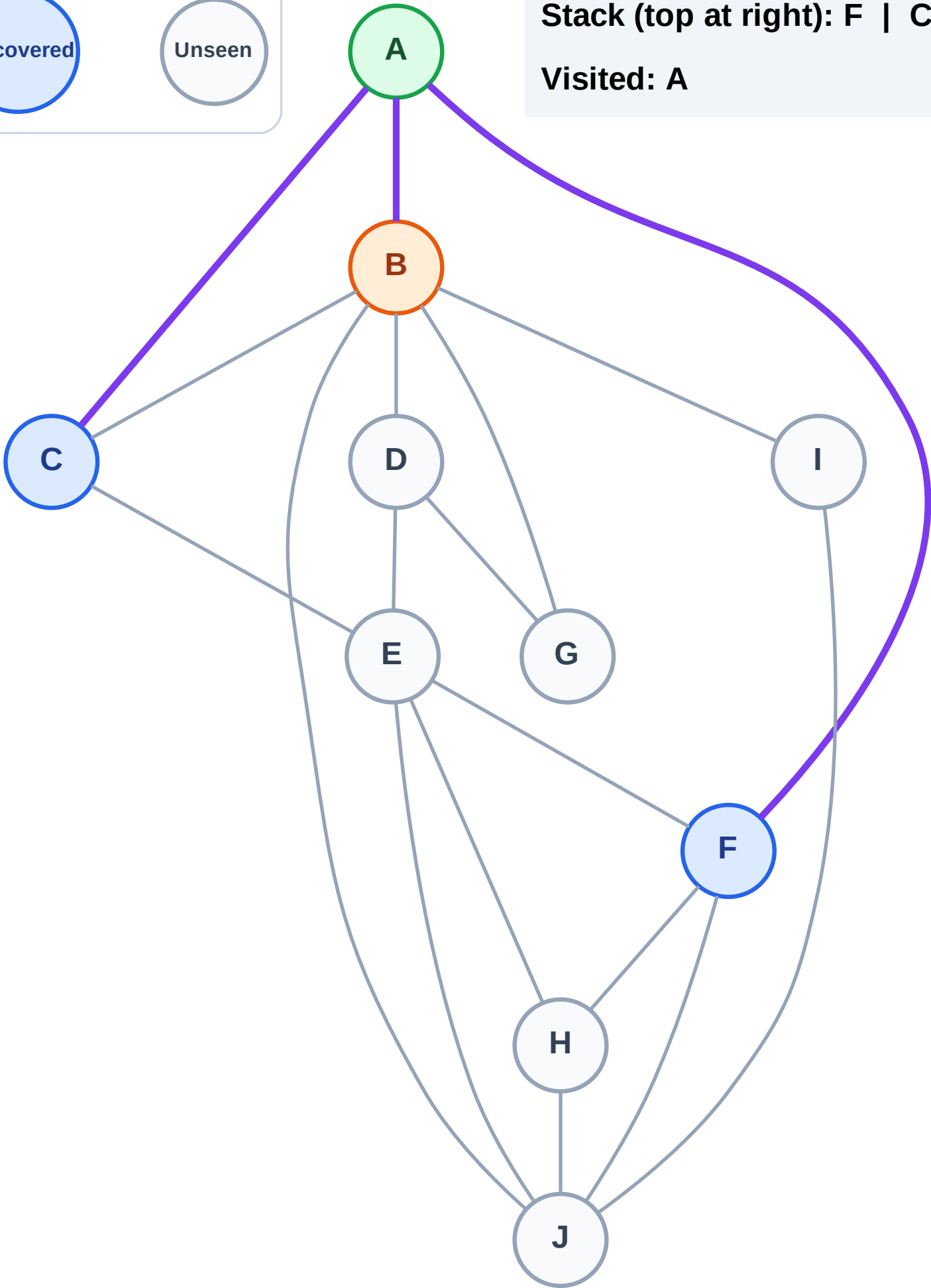
Complete

Processing

Discovered

Unseen

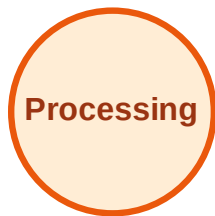
Stack (top at right): F | C
Visited: A



DFS Traversal

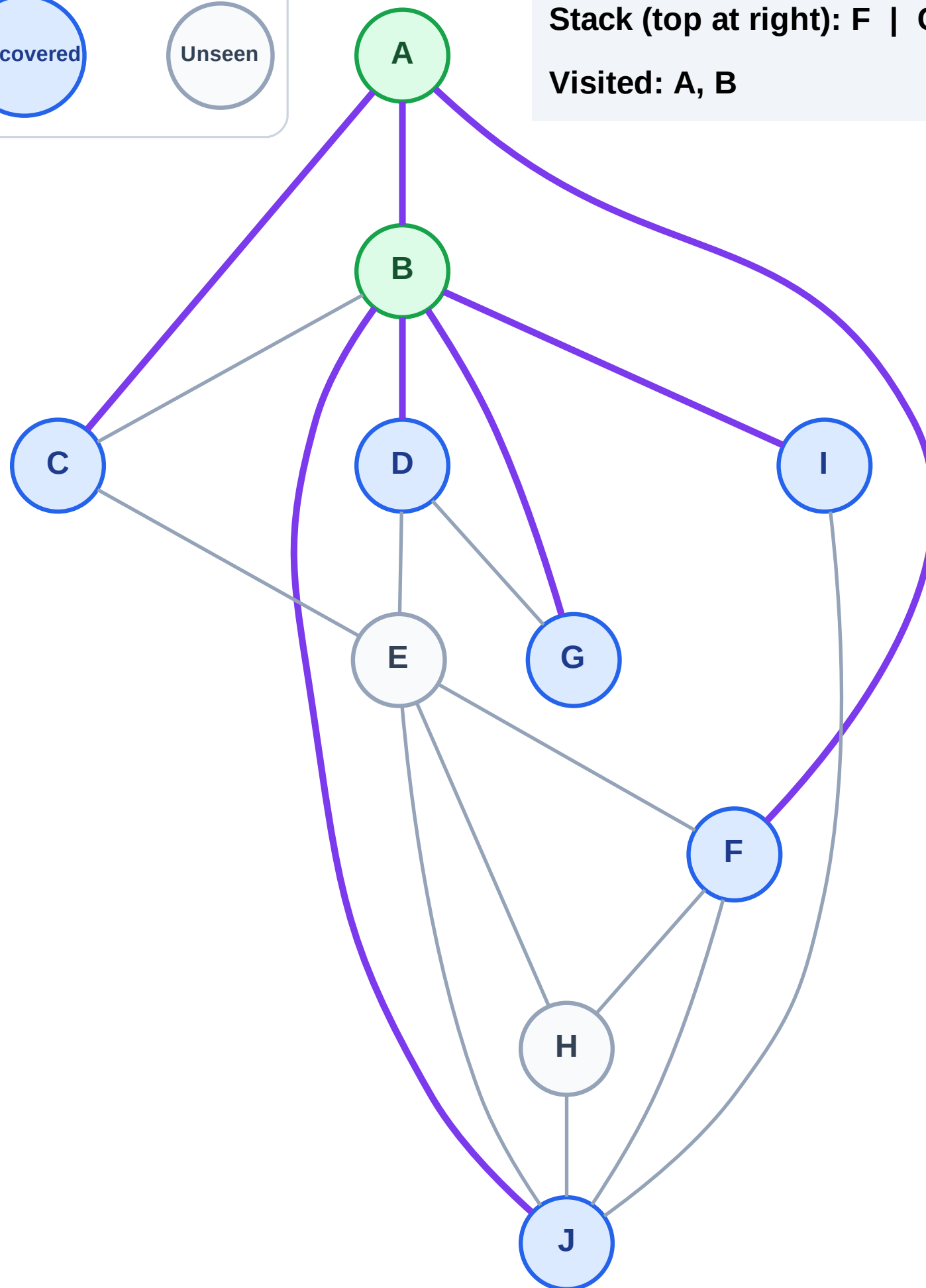
Step 5: Discover J, I, G, D from B

Legend



Stack (top at right): F | C | J | I | G | D

Visited: A, B



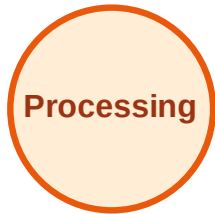
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



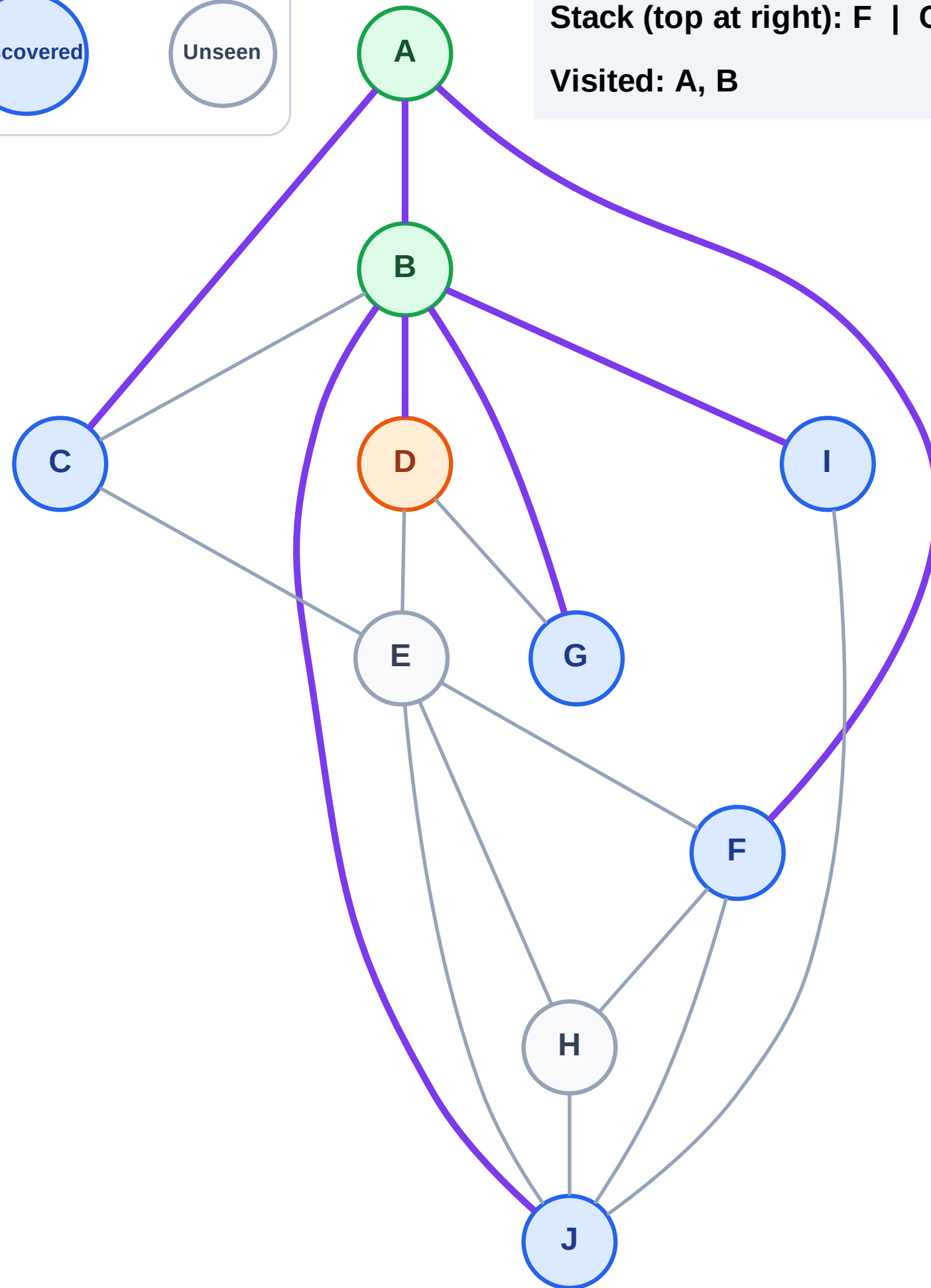
Discovered



Unseen

Stack (top at right): F | C | J | I | G

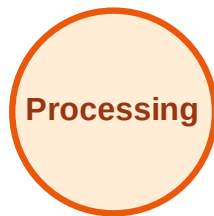
Visited: A, B



DFS Traversal

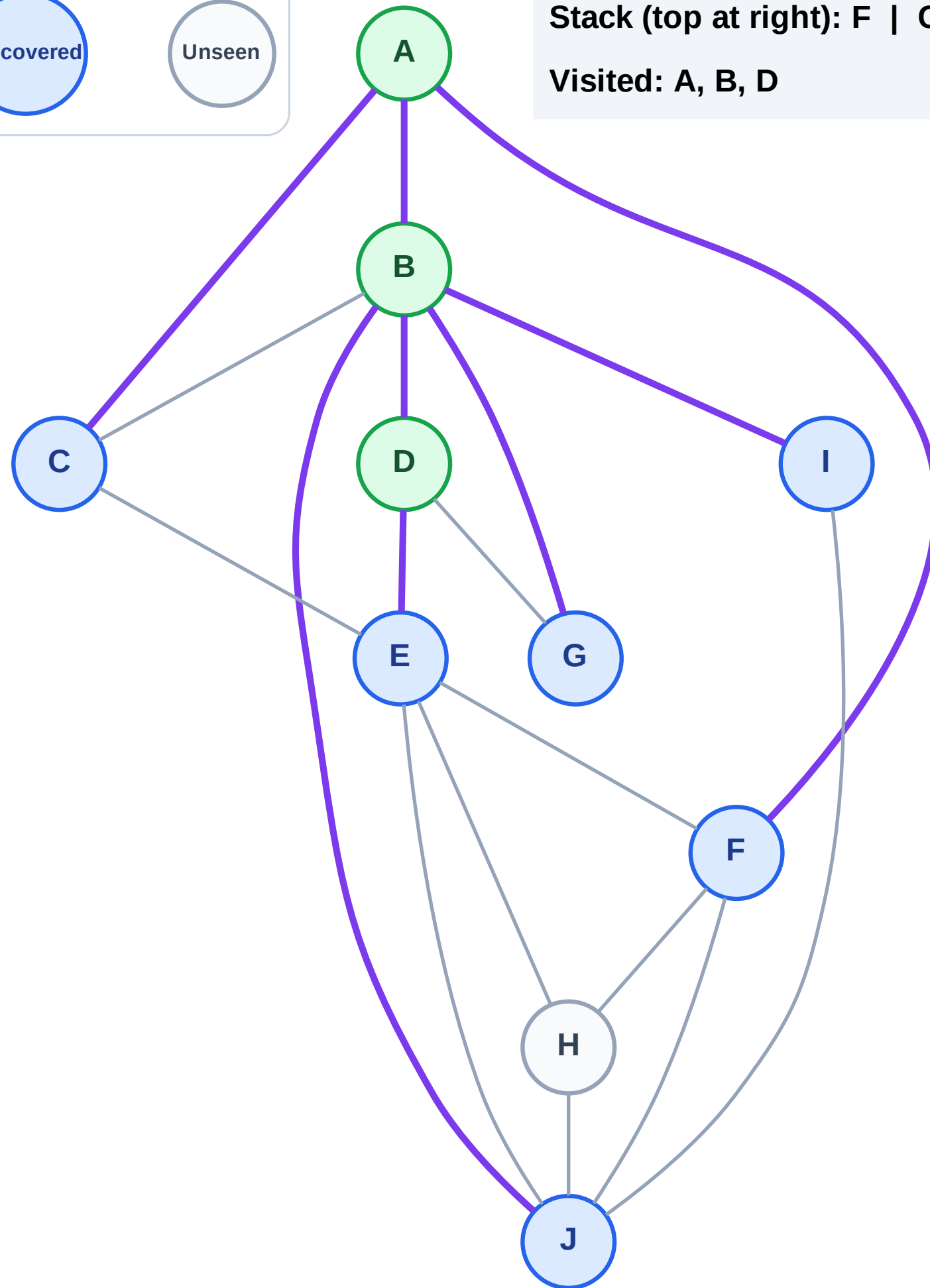
Step 7: Discover E from D

Legend



Stack (top at right): F | C | J | I | G | E

Visited: A, B, D



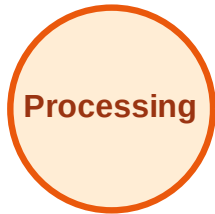
DFS Traversal

Step 8: Process node E

Legend



Complete



Processing



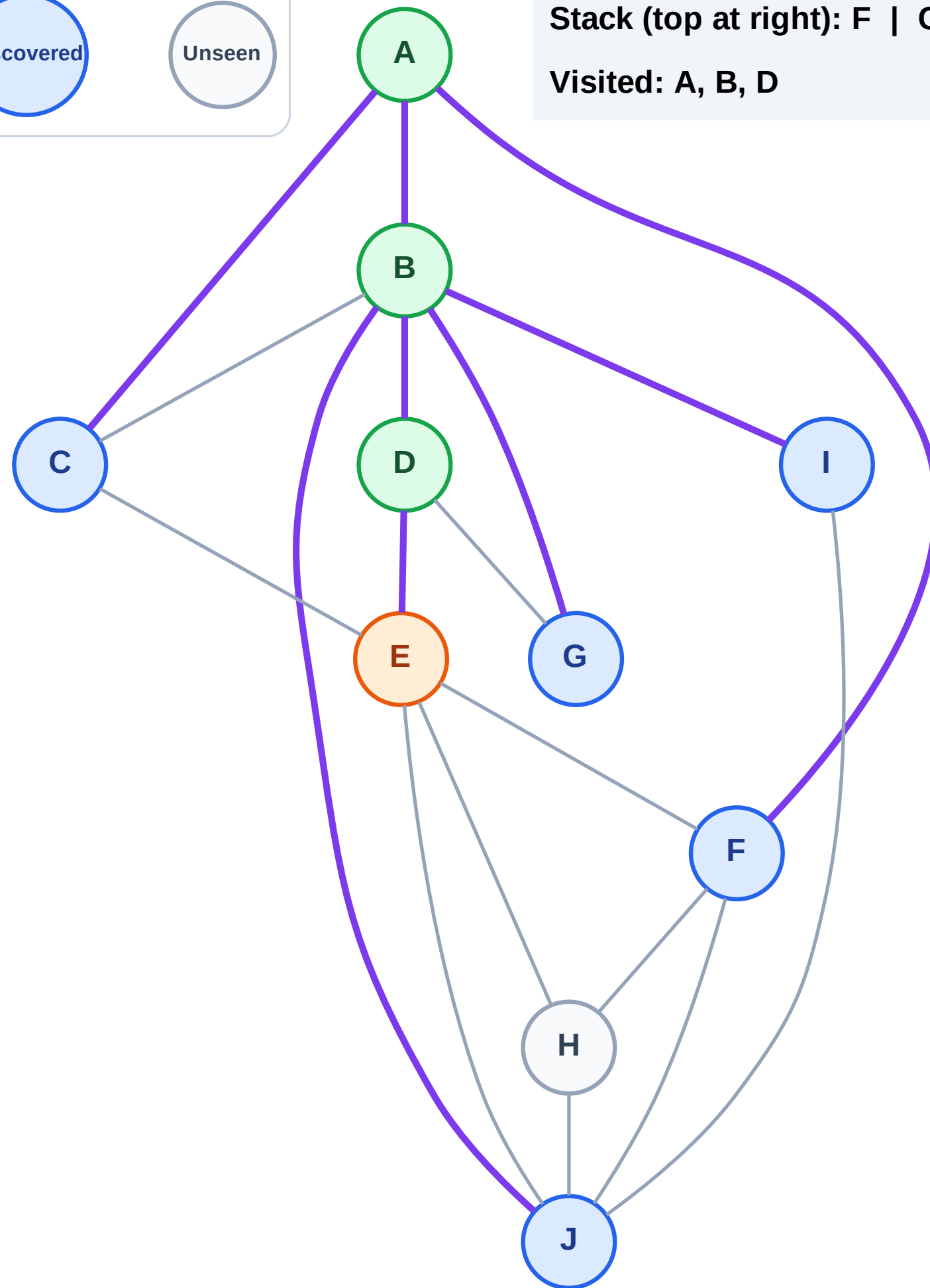
Discovered



Unseen

Stack (top at right): F | C | J | I | G

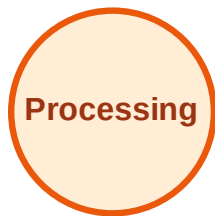
Visited: A, B, D



DFS Traversal

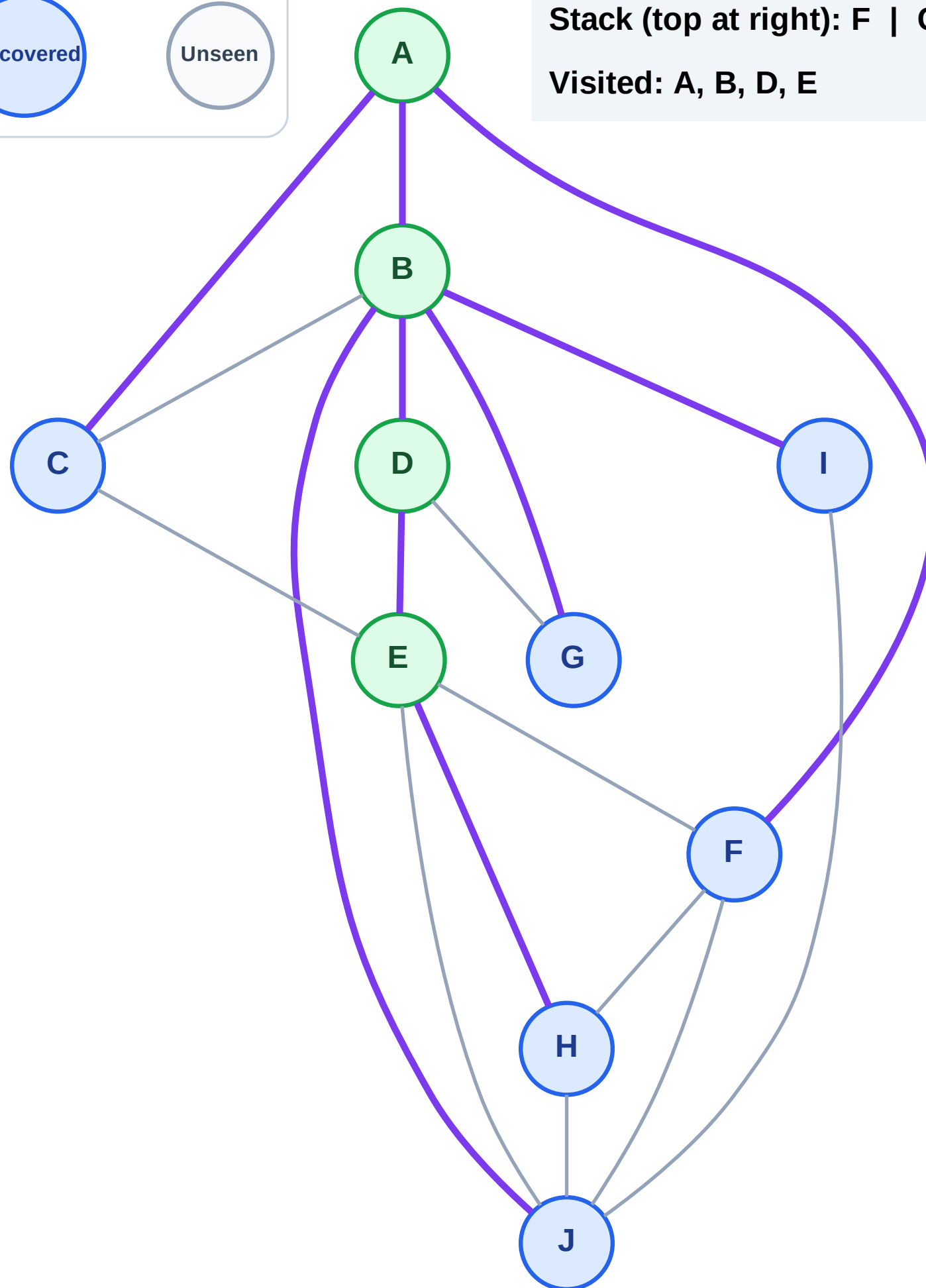
Step 9: Discover H from E

Legend



Stack (top at right): F | C | J | I | G | H

Visited: A, B, D, E



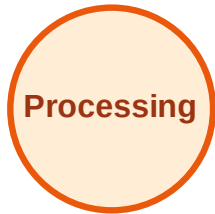
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



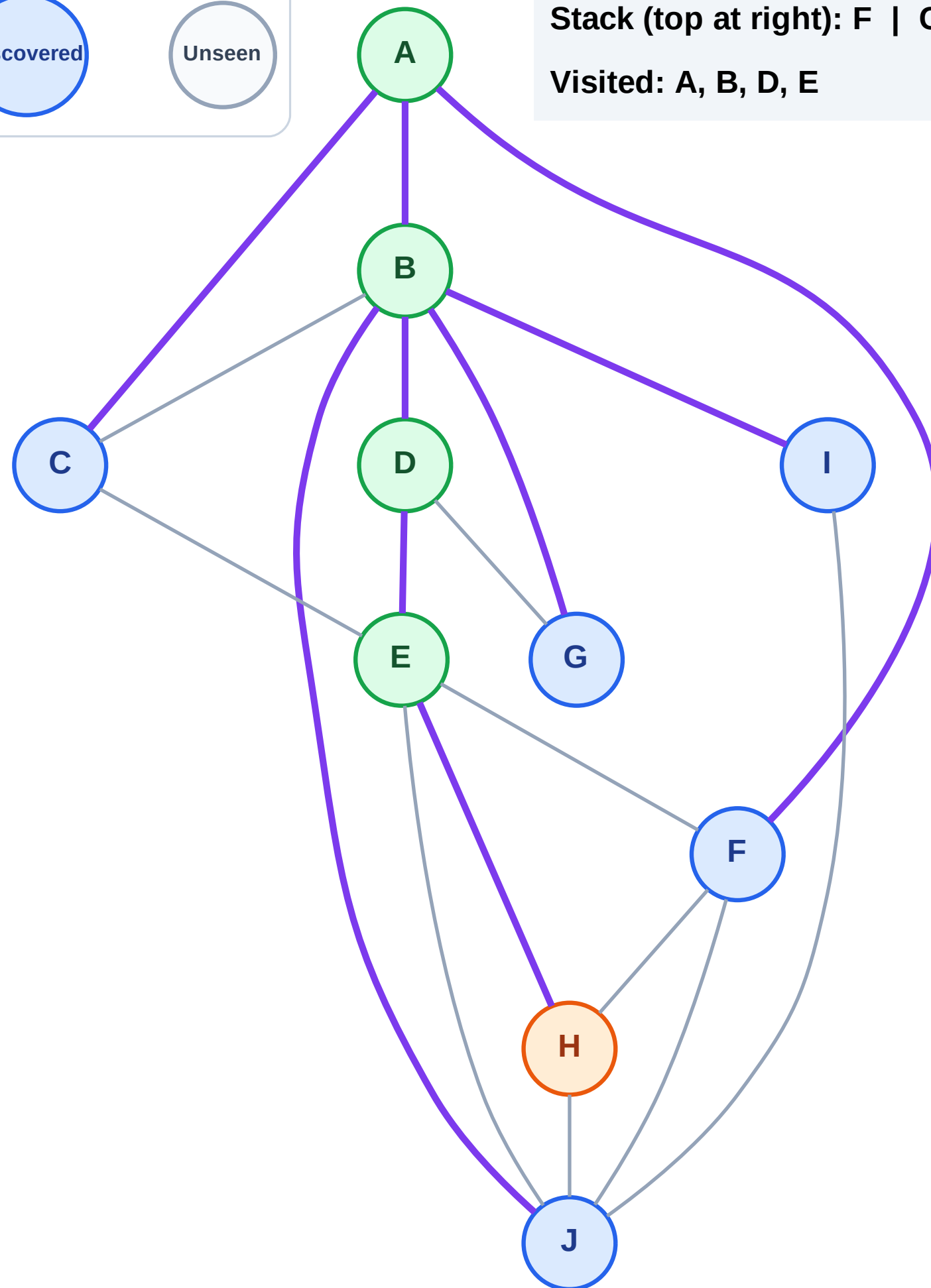
Discovered



Unseen

Stack (top at right): F | C | J | I | G

Visited: A, B, D, E



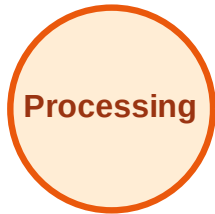
DFS Traversal

Step 11: No new neighbors from H

Legend



Complete



Processing



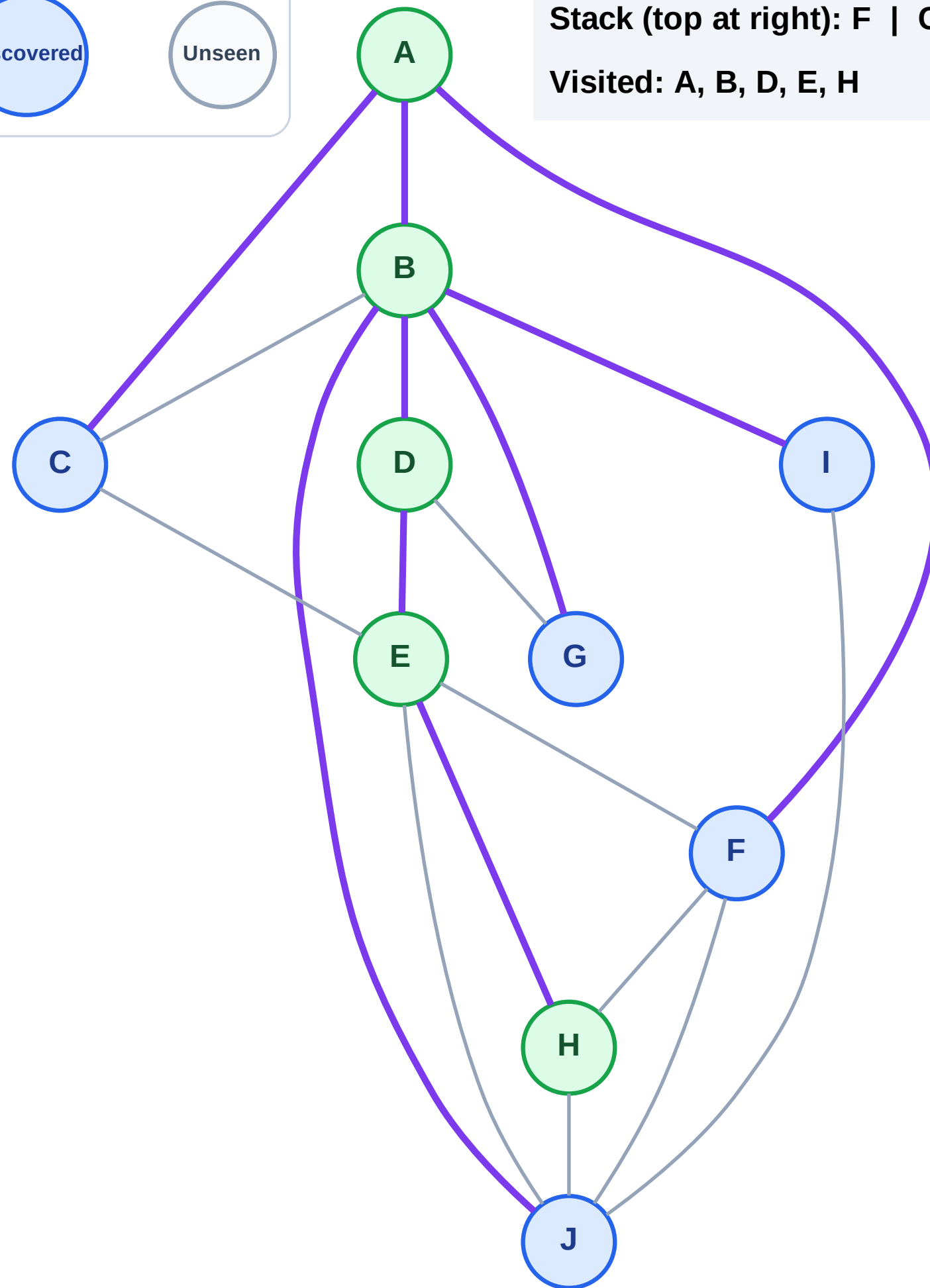
Discovered



Unseen

Stack (top at right): F | C | J | I | G

Visited: A, B, D, E, H



DFS Traversal

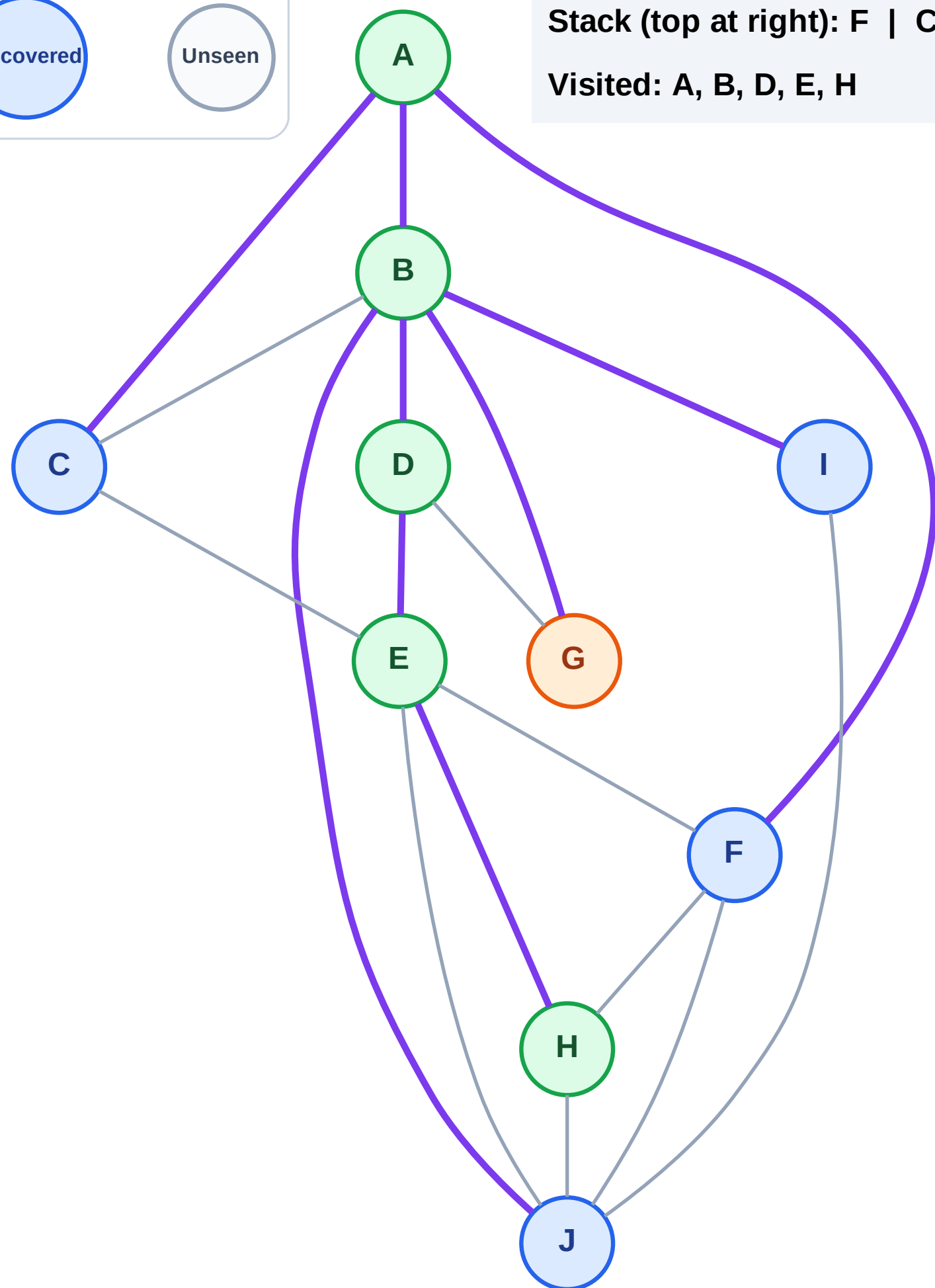
Step 12: Process node G

Legend



Stack (top at right): F | C | J | I

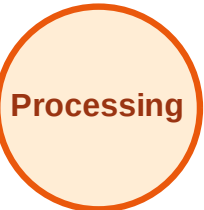
Visited: A, B, D, E, H



DFS Traversal

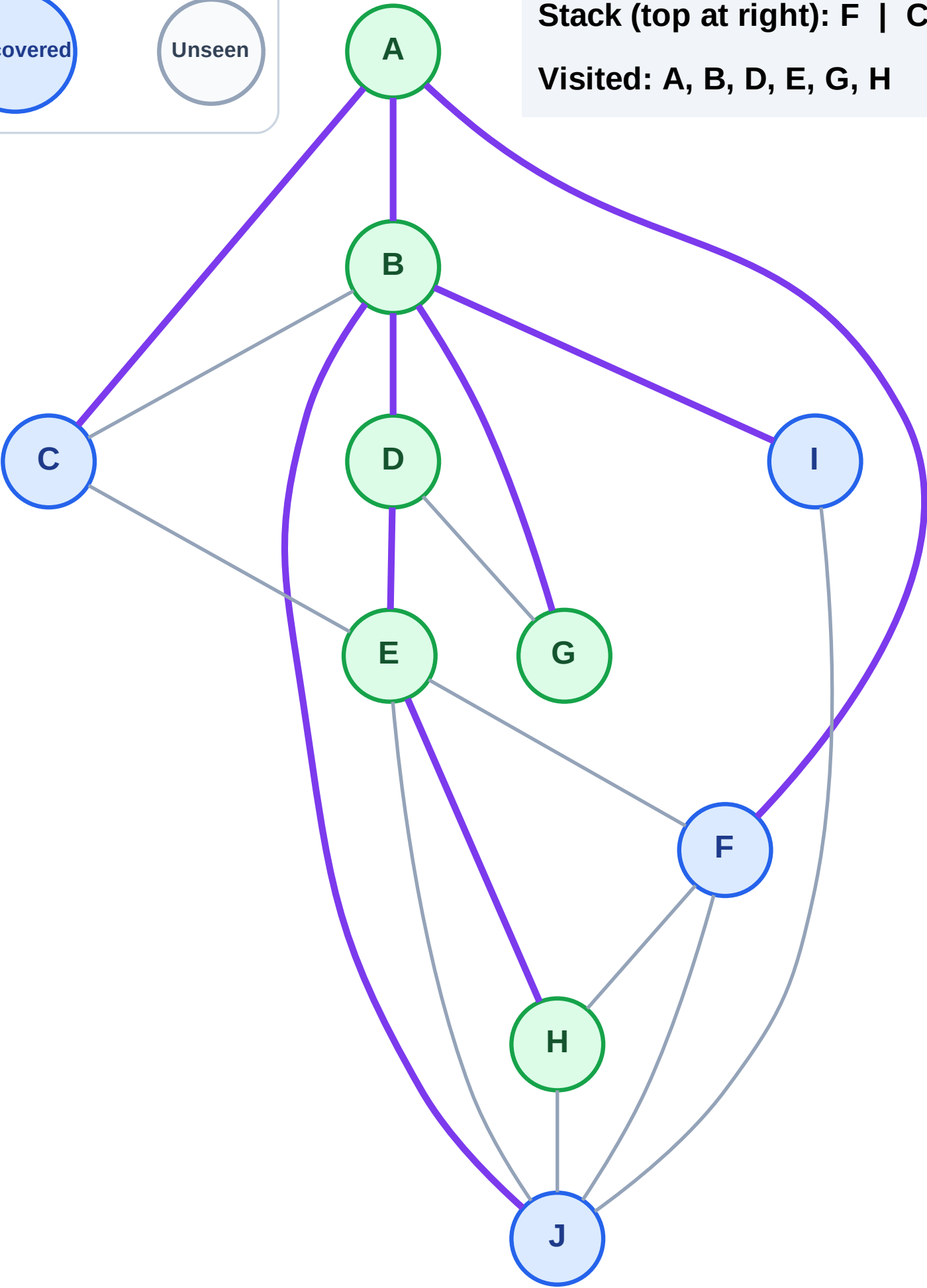
Step 13: No new neighbors from G

Legend



Stack (top at right): F | C | J | I

Visited: A, B, D, E, G, H



DFS Traversal

Step 14: Process node I

Legend

Complete

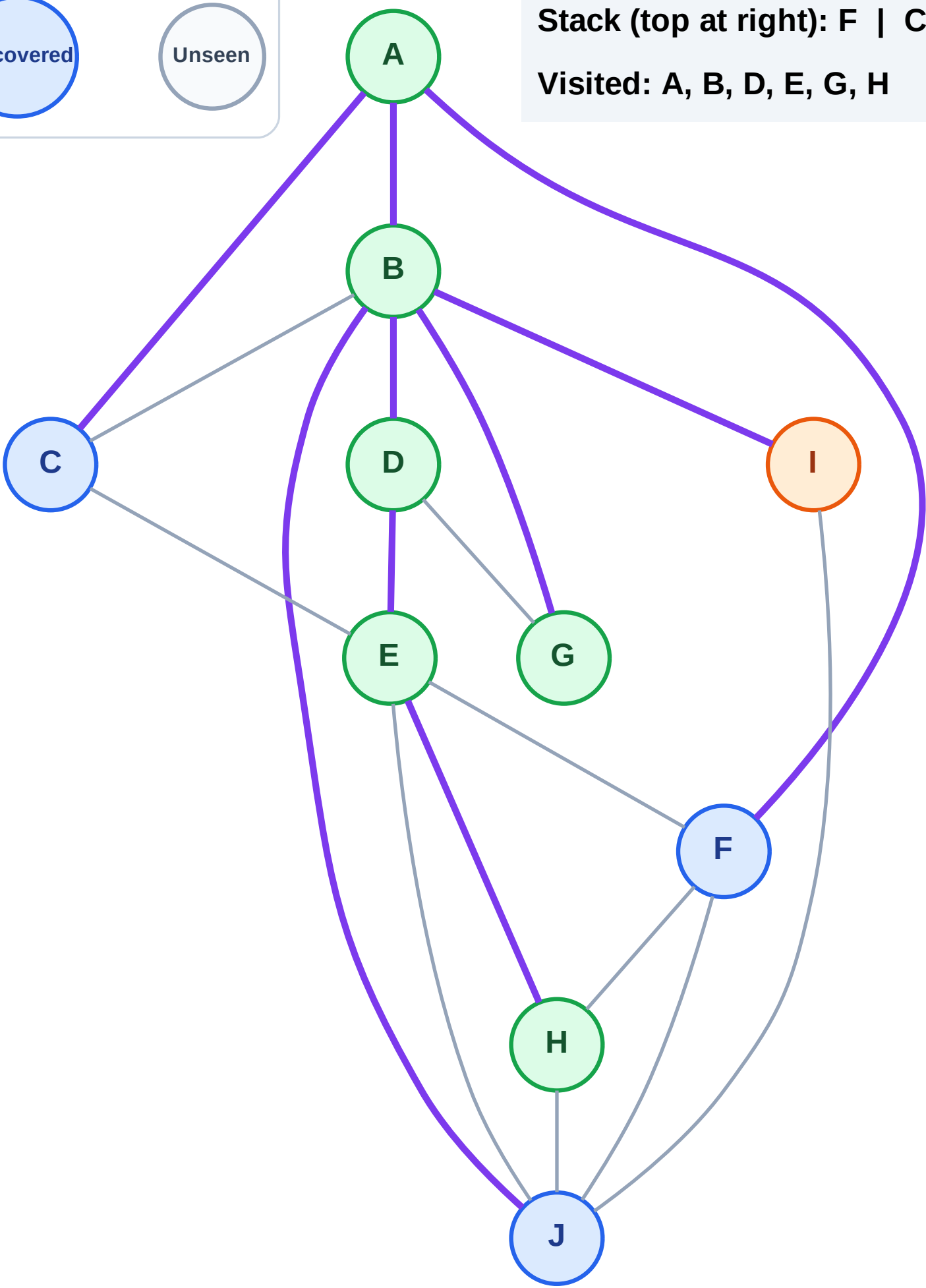
Processing

Discovered

Unseen

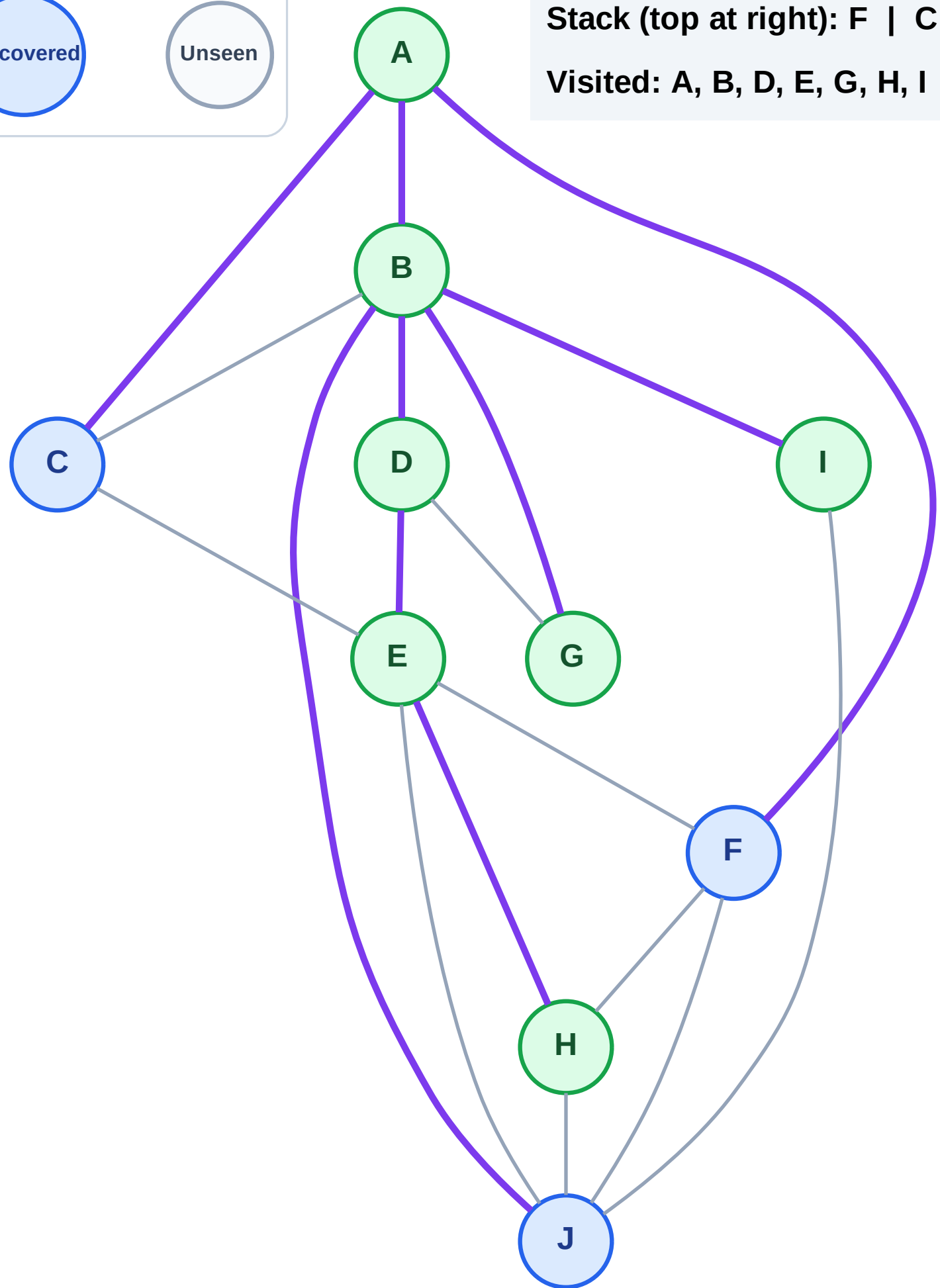
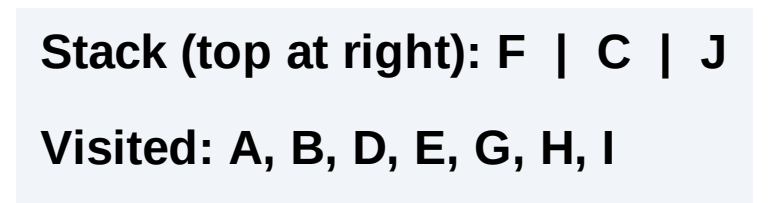
Stack (top at right): F | C | J

Visited: A, B, D, E, G, H



Step 15: No new neighbors from I

Step 15: No new neighbors from I



DFS Traversal

Step 16: Process node J

Legend

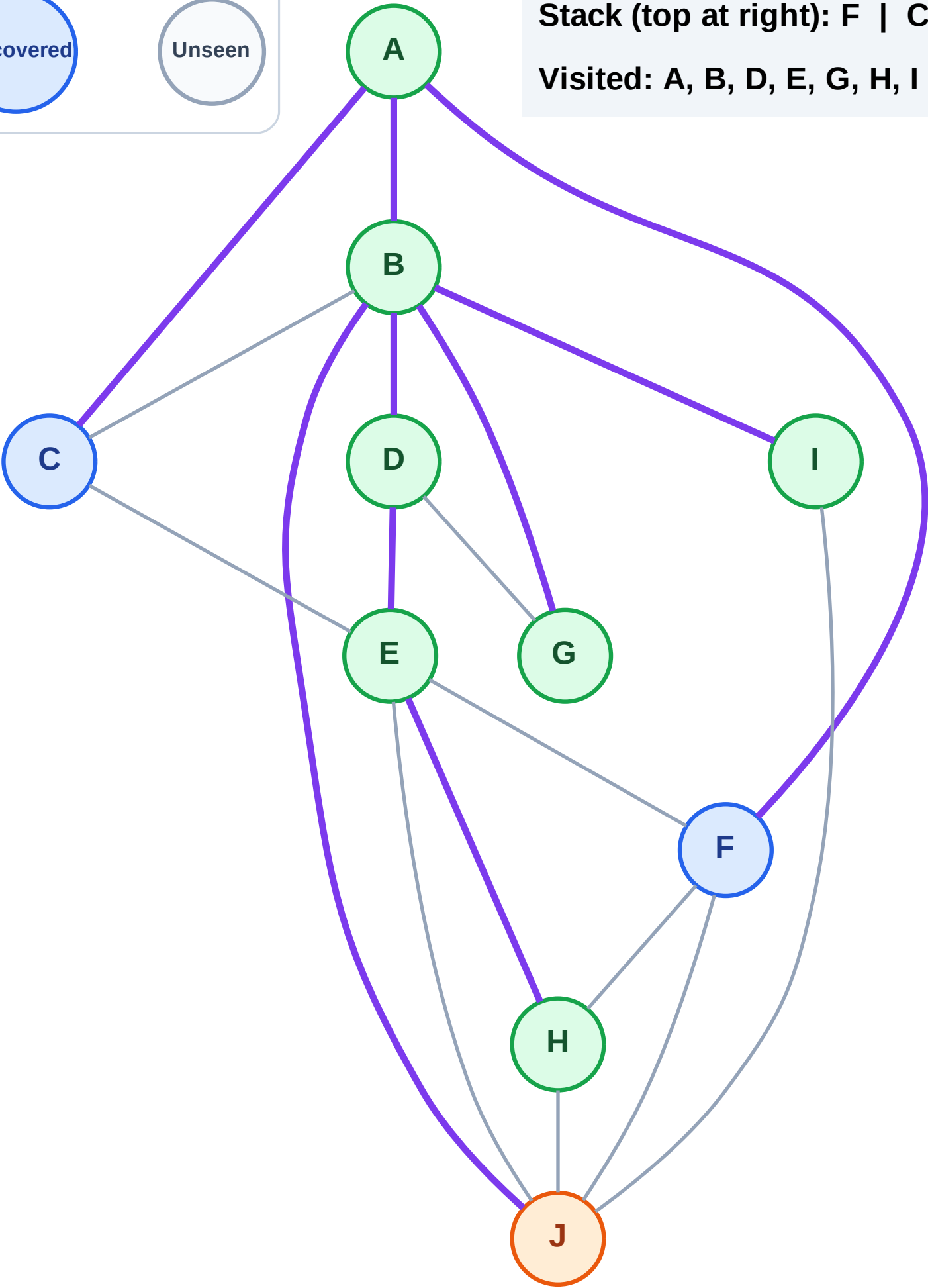
Complete

Processing

Discovered

Unseen

Stack (top at right): F | C
Visited: A, B, D, E, G, H, I



DFS Traversal

Step 17: No new neighbors from J

Legend

Complete

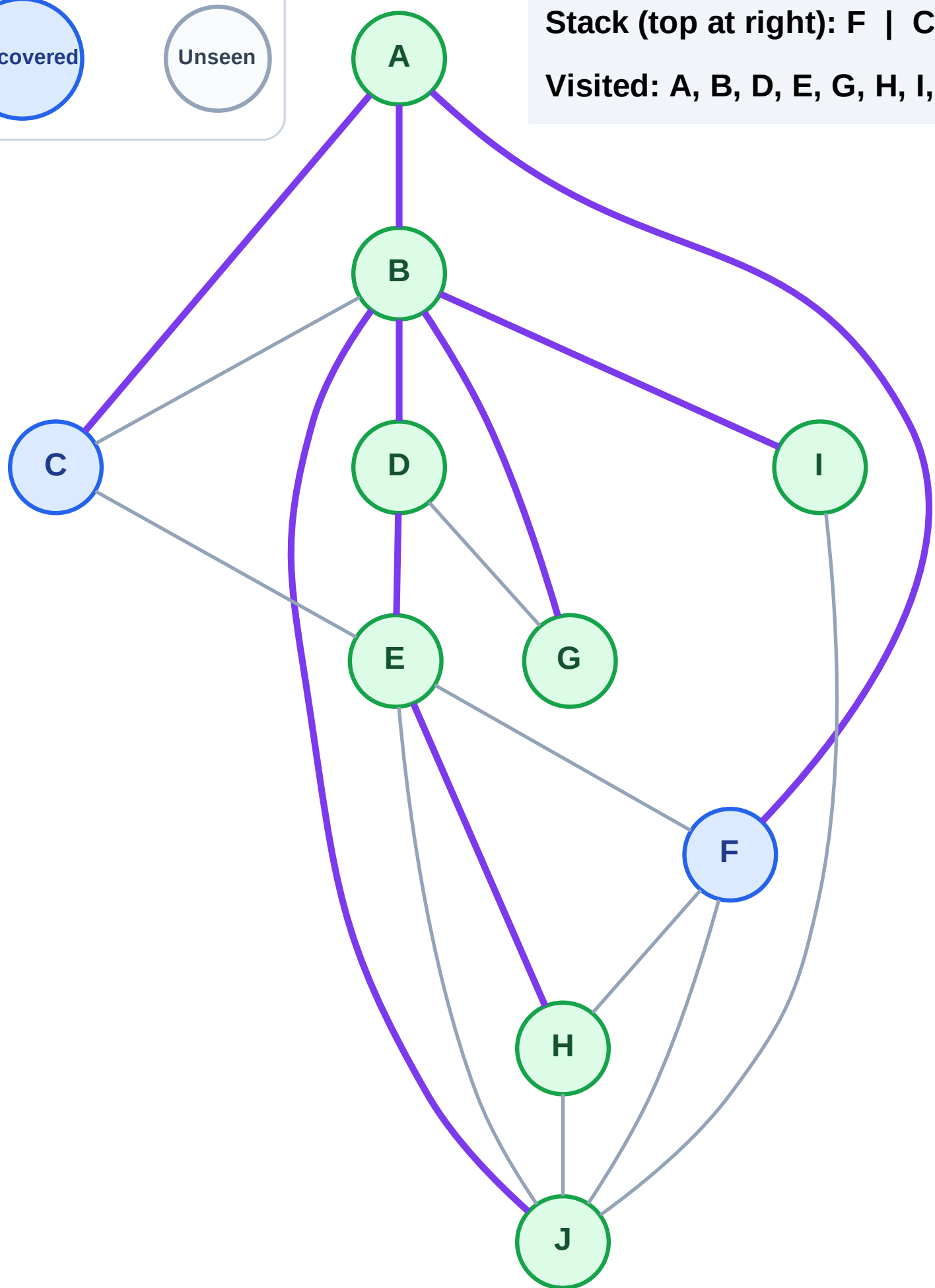
Processing

Discovered

Unseen

Stack (top at right): F | C

Visited: A, B, D, E, G, H, I, J



DFS Traversal

Step 18: Process node C

Legend

Complete

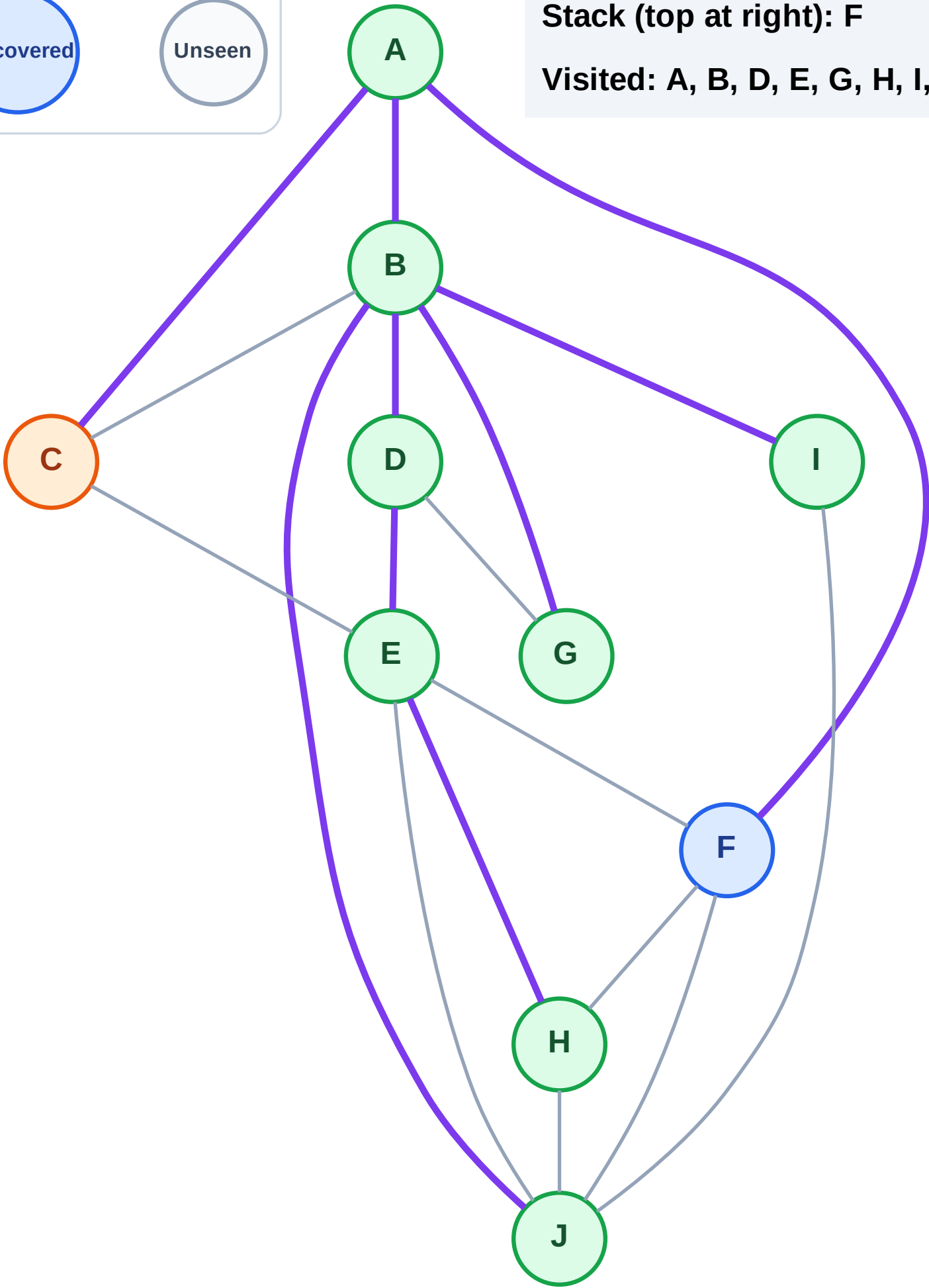
Processing

Discovered

Unseen

Stack (top at right): F

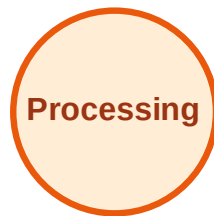
Visited: A, B, D, E, G, H, I, J



DFS Traversal

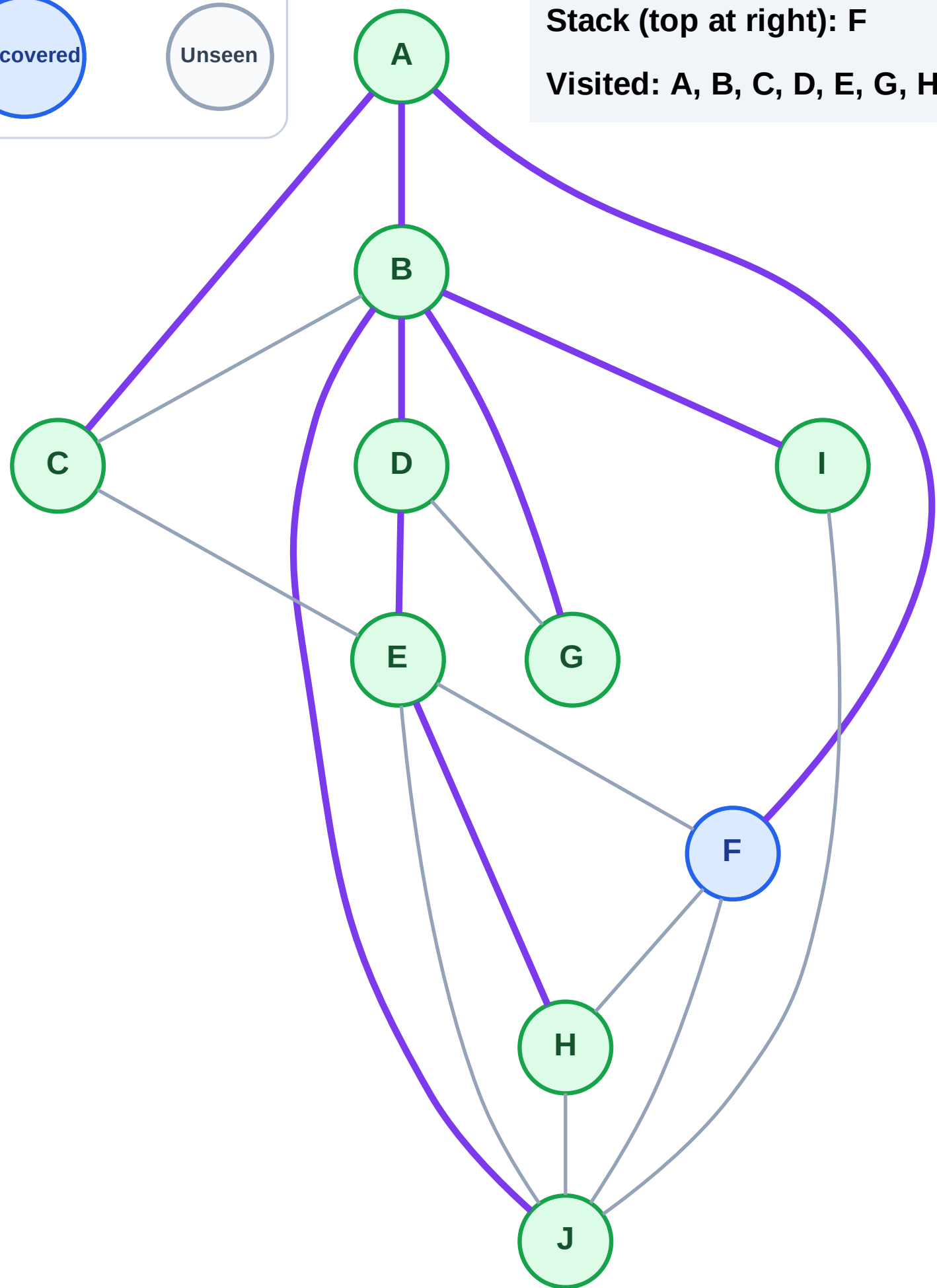
Step 19: No new neighbors from C

Legend



Stack (top at right): F

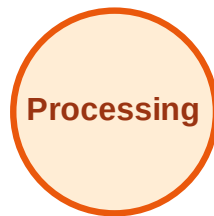
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

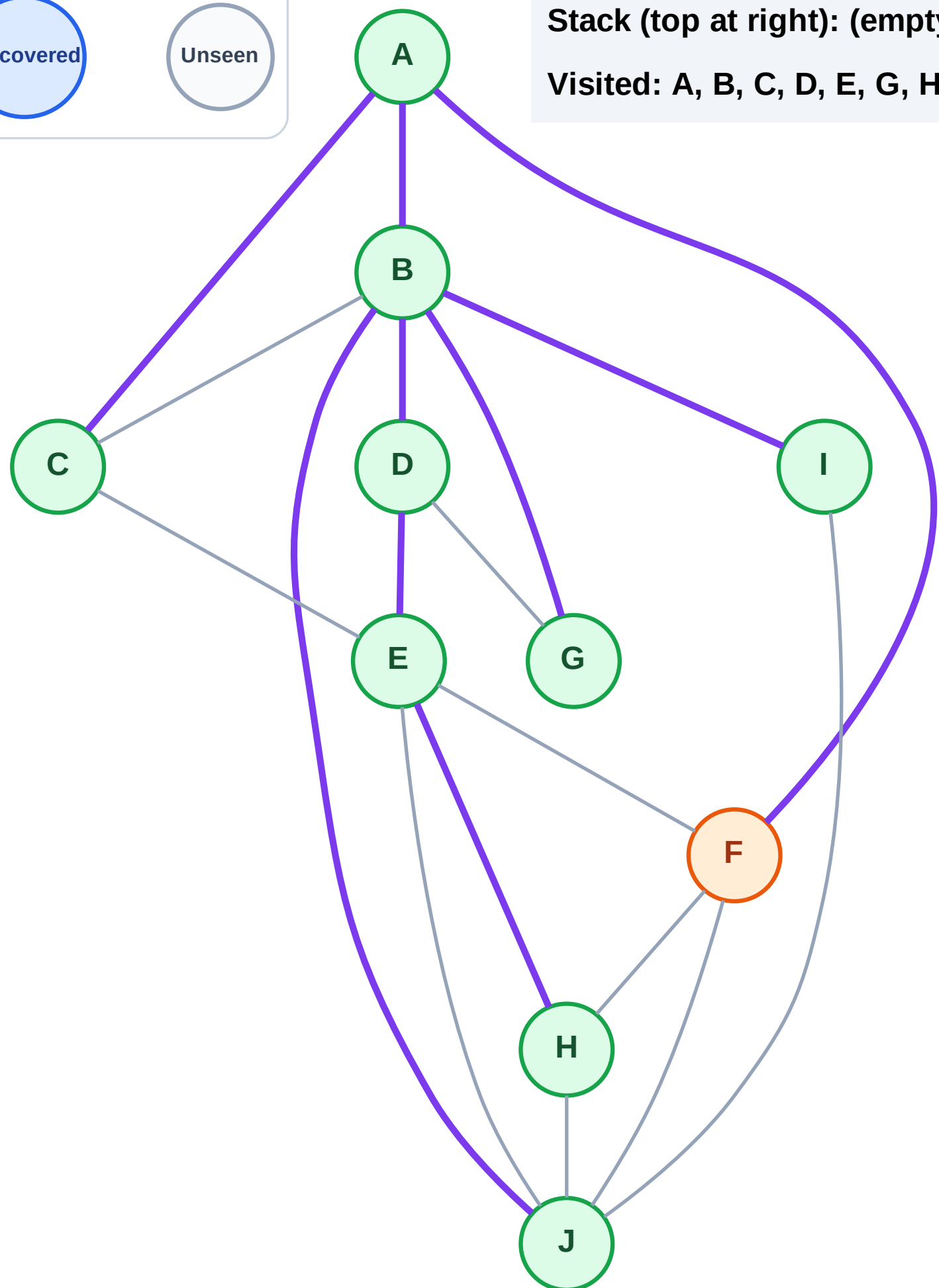
Step 20: Process node F

Legend



Stack (top at right): (empty)

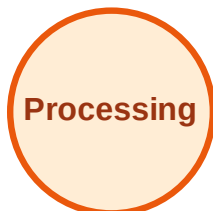
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

